Contents

1 Point Estimation

Definition: A point estimator is any scalar (or vector) -valued function of the sample. $(x_1, ...x_n) \sim f(x|\theta)$

A point estimator for $\tau(\theta)$ is a statistic $T(x)$ with the purpose of approximating $\tau(\theta)$

1.1 Method of Moments

The k-th moment of a r.v. X is $\mu_k(\theta) = \mathbb{E}_\theta(X^k) = \int\limits_{\mathbb{X}} x^k f(x|\theta)dx$

Given an iid sample $x_1...x_n \overset{\text{iid}}{\sim} f(x|\theta)$ we have sample moments: $\hat{\mu}_k = \frac{1}{n}\sum\limits_{i=1}^{n} x_i^k$

Suppose $\theta \in \Theta \in \mathbb{R}^P$, and that $\mu_k(\theta)$ exists and is fininte for k = 1, ..., p.

Definition: The method of moments estimator of θ is the solution to the system of equations:

$$\mu_1(\theta) = \hat{\mu}_1 \tag{1}$$

$$. \tag{2}$$

$$. \tag{3}$$

$$\mu_p(\theta) = \hat{\mu}_p \tag{4}$$

We call it $\hat{\theta}_{MM}$

Example: $X_i \overset{\text{iid}}{\sim} Beta(\alpha, \beta), \theta = (\alpha, \beta)$

$\mu_1(\theta) = \frac{\alpha}{\alpha+\beta}$

$\mu_2(\theta) = Var_\theta(x_1) + \left(\frac{\alpha}{\alpha+\beta}\right)^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} + \frac{\alpha^2}{(\alpha+\beta)^2}$

$\frac{\alpha}{\alpha+\beta} = \hat{\mu}_1, \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} + \frac{\alpha^2}{(\alpha+\beta)^2} = \hat{\mu}_2$

$\beta = \left(\frac{1-\hat{\mu}_1}{\hat{\mu}_1}\right)\alpha$ from the first equation. Plug this into the second expression and solve:

$\hat{\alpha}_{MM} = \hat{\mu}_1 \left[\frac{\hat{\mu}_1(1-\hat{\mu}_1)}{\hat{\mu}_2-\hat{\mu}_1^2} - 1\right]$

$\implies \hat{\beta}_{MM} = \frac{(1-\hat{\mu}_1)}{\hat{\mu}_1}\hat{\alpha}_{MM}$

Example: $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

$\mu_1(\theta) = \hat{\mu}_1, \mu_1(\theta) = \mu, \implies \hat{\mu}_{MM} = \mu$

$\mu_2(\theta) = Var_\theta(x_1) + \mu_1^2 = \sigma^2 + \mu^2 = \hat{\mu}_2$

$\sigma_{MM}^2 = \hat{\mu}_2 - \hat{\mu}_1^2 = \frac{n-1}{n}s^2$

Note: If μ, σ^2 are the mean/variance of any family, their MM estimators are $\overline{X}$, $\frac{n-1}{n}s^2$.

For families where parameters are not just the mean and variance, you can find it via two ways: use mean/variance in MM, then calculate parameters, or use parameters in MM then calculate mean/variance. Both yield same results.

Fact: MM estimators are invariant of re-parameterizations.

Let $\eta = \eta(\theta)$ be a 1:1 mapping (intervtible). Then, $\hat{\eta}_{MM} = \eta(\hat{\theta}_{MM})$

Let's say $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ and we want to estimate $\tau(\theta) = \frac{\mu}{\sigma}$. This isn't 1:1. What can we do? We can do the 'Transformations' method and create a second value τ_2, etc. We can also just plug in the estimators.

Definition: The MM estimator for a parametric function $\tau(\theta)$ is just $\hat{\tau}_{MM}(\theta) = \tau(\hat{(\theta)}_{MM})$

Properties:

- MM equations may have a unique solution, no solution, or many solutions
- Often, MM estimators are used as initial values for another estimation technique (ie. a root finding method)
- Why should it work? Let's say θ^* is the true value of θ. Then Law of Large Numbers says: $\hat{\mu}_k \xrightarrow{P} \mu_k(\theta*)$. Then we are solving $\mu_k(\theta) = \hat{\mu}_k \approx \mu_k(\theta^*)$

Example: $X_i \overset{\text{iid}}{\sim} Bin(m, \theta), i = 1, ..., n$ (m is known)

Find the MM estimator of $\tau(\theta) = ln\frac{\theta}{1-\theta}$

1. Find MM for θ ($\hat{\theta}_{MM}$)
2. Plug in ($\hat{\tau}_{MM}(\theta) = \tau(\hat{\theta}_{MM})$)
3. $\hat{\tau}_{MM}(\theta) = ln\frac{\overline{X}/m}{1-\overline{X}/m}$

2 Maximum Likelihood

$X = (X_1, ..., X_n) \sim f(x|\theta), \theta \in \Theta \subset \mathbb{R}^k$

Notation: $f(x;\theta)$: function of x indexed at θ. Basically, given some set value of the θ.

Likelihood Function: The likelihood function is: $L(\theta; x) = f(x|\theta)$

Notes:

- L is a function of θ for each $x \in \mathbb{X}$ (in sample space)
- Plugging in X for x gives $L(\theta; X)$, a stochastic process (ie. plug in a random X makes this a random function for θ)
- The log-likelihood function is $l(\theta; x) = ln[L(\theta; x)]$
- If $x_i \overset{\text{iid}}{\sim} f(x_i|\theta)$ (f is marginal dist.), then $l(\theta; x) = \sum_{i=1}^{n} ln[f(x_i|\theta)]$ (because $x_i \overset{\text{iid}}{\sim}$, the sum is just a transformation and we can apply LLN, CLT, etc.)

Maximum Likelihood Estimate: If $x \in \mathbb{X}$ is observed, a maximum likelihood estimate of $\theta, \hat{\theta}(x)$, is any value $\theta \in \Theta$ that maximizes $L(\theta|x)$.

$\hat{\theta}(x) = \underset{\theta \in \Theta}{argmax}[L(\theta|x)]$

This is a function of observed data (an estimate, not an estimator).

Maximum Likelihood Estimator: A maximum likelihood estimator (MLE) is $\hat{\theta} = \hat{\theta}(X)$

If an ML estimate exists, then $\hat{\theta}(x) = \underset{\theta \in \Theta}{argmax}[l(\theta; x)]$. This is because $ln(x)$ is a strictly increasing function.

Why does maximum likelihood work? Can we show that $\hat{\theta} \approx \theta_0$ (true parameter)?

Assume $X_1, ..., X_n \overset{\text{iid}}{\sim} f(x_i|\theta)$. $l(\theta|x) = \sum_{i=1}^{n} ln[f(x_i|\theta)]$

$\frac{1}{n} l(\theta|x) = \frac{1}{n} \sum_{i=1}^{n} ln[f(x_i|\theta)] \xrightarrow{P} \mathbb{E}_{\theta_0}(ln[f(x|\theta)]) = \int_{\mathbb{X}} ln[f(x|\theta)] f(x|\theta_0) dx$

It would make sense that $\hat{\theta}(x) = \underset{\theta \in \Theta}{argmax}[l(X|\theta)] \approx \underset{\theta \in \Theta}{argmax}\, \mathbb{E}_{\theta_0}(ln[f(X|\theta)])$ which we hope $= \theta_0$.

Define $D(\theta; \theta_0) = \mathbb{E}_{\theta_0}(ln[f(X|\theta)])$. We will show that $D(\theta_0; \theta_0) - D(\theta; \theta_0) \geq 0 \forall \theta$.

Kullback-Liebler Divergence: Let f_0 and f_1 be any two PDFs/PMFs. The Kullback-Liebler divergence from f_0 to f_1 is $K(f_0, f_1) = -\mathbb{E}_{f_0}[ln \frac{f_1(x)}{f_0(x)}]$

$D(\theta_0; \theta_0) - D(\theta; \theta_0) = \mathbb{E}_{\theta_0}[ln(f(X|\theta_0)) - ln(f(X|\theta))] = -\mathbb{E}_{\theta_0}[ln \frac{f(X|\theta)}{f(X|\theta_0)}]$.

Lemma: For any two PDFs/PMFs $f_0, f_1, K(f_0, f_1) \geq 0$, with equality iff $f_0 \equiv f_1$.

Remeber Jensen's Inequality: When $g(x)$ is convex (happy), $\mathbb{E}[g(x)] \geq g(\mathbb{E}[x])$.

Proof (Discrete Case): Suppose $X \sim f_0$ and set $Z = \frac{f_1(x)}{f_0(x)}$. Let $S_j = \{x : f_j(x) > 0\}$

Since $g(z) = -ln(z)$ is convex and $\mathbb{E}_{f_0}(z) = \sum\limits_{x \in S_0} \frac{f_1(x)}{f_0(x)} f_0(x) = \sum\limits_{x \in S_0} f_1(x) \leq 1$.

By Jensen's Inequality: $K(f_0, f_1) = -\mathbb{E}_{f_0}[ln(Z)] = \mathbb{E}_{f_0}[g(z)] \underset{1}{\geq} g(\mathbb{E}_{f_0}(z)) \underset{2}{\geq} 0$.

This is only 'equal' when g is linear. Since $g(z)$ is not linear, equality in 1 only happens iff $Z = \frac{f_1(x)}{f_0} = c \neq 0, \forall x \in S_0, [S_0 \subset S_1]$. Equality in 2 only happens iff $\sum\limits_{x \in S_0} f_1(x) = 1, [S_1 \subset S_0]$.

Suppose 1 and 2 are equalities. $1 = \sum\limits_{x \in S_1} f_1(x) = \sum\limits_{x \in S_1} c f_0(x) = c \sum\limits_{x \in S_1} f_0(x) = c \sum\limits_{x \in S_0} f_0(x) = c$

3 Decision Theory

3.1 Bayesian Estimation

$x_1, ..., x_n \sim f(x|\theta), \theta \in \Theta$

Definition: A priori distribution π for θ is a PDF/PMf over Θ: $\int\limits_{\Theta} \pi(\theta)d\theta = 1, \pi(\theta) \geq 0, \theta \in \Theta$

Main Idea: π tells us what θs are "important". $\pi(\theta|x)$ tells us which are important afetr knowing x.

Definition: If we observed $x \in X$, the posterior distribution is: $\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int\limits_{\Theta} f(\theta|x)\pi(\theta)d\theta}$

Here, $m(x) = \int\limits_{\Theta} f(\theta|x)\pi(\theta)d\theta$ is the "marginal" distirbution of x.

Some logical things to do with $\pi(\theta|x)$:

1. Estimate θ using a measure of 'center':
2. $\hat{\theta} = \mathbb{E}(\theta|x) = \int\limits_{\Theta} \theta\pi(\theta|x)d\theta$
3. $0.5 = \int\limits_{0}^{\hat{\theta}} pi(\theta|x)d\theta$
4. $\hat{\theta}(x) = \underset{\theta\in\Theta}{argmax}[\pi(\theta|x)]$

Example: $x_i \overset{\text{iid}}{\sim} Bern(\theta), \theta \sim Beta(\alpha, \beta)$ (the common prior for the Bernoulli is the Beta: also defined on 0:1)

$$\begin{aligned}\pi(\theta) &= \frac{1}{\beta(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}, 0 < \theta < 1 \\ &\propto \theta^{\alpha-1}(1-\theta)^{\beta-1}\end{aligned}$$

$f(x|\theta) = \prod\limits_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}$

$\pi(\theta|x) = \frac{\theta^{\sum x_i}(1-\theta)^{n-\sum x_i}\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)\int\limits_{0}^{1} \frac{\theta^{\sum x_i}(1-\theta)^{n-\sum x_i}\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}d\theta}$

Denominator is just a function of x (integrating out θ) and some constants.

$\implies \pi(\theta|x) \propto \theta^{\alpha+\sum x_i-1}(1-\theta)^{\beta+n-\sum x_i-1}$

i.e., $\theta|x \sim Beta(\alpha + \sum x_i, \beta + n - \sum x_i)$

The posterior mean is $\hat{\theta}(x) = \frac{\alpha+\sum x_i}{\alpha+\beta+n}$

Take-aways from this example:

1. Both $\pi(\theta)$ and $\pi(\theta|x)$ were Beta distributions. We say then that the Beta is conjugate for the Bernoulli likelihood
2. We did not need to compute the $m(x)$. In practice, usually can't compute it and instead rely on smapling (MCMC, etc.)

3. The usual frequentist estimate is $\bar{x}$ (this would be the limiting case of $\alpha = \beta = 0$). In fact, $\hat{\theta}_B(x) = \frac{\alpha}{\alpha+\beta+n} + \frac{\sum x_i}{\alpha+\beta+n} = \left(\frac{\alpha}{\alpha+\beta}\right)\left(\frac{\alpha+\beta}{\alpha+\beta+n}\right) + \bar{X}\left(\frac{n}{\alpha+\beta+n}\right)$. That is, the posterior mean is a weighted average of the prior mean and the MLE. As α or β go to infinity, result is dominated by prior. As n goes to infitinty, get more weight on the $\bar{X}$

3.2 Conjugate Priors

Definition: Let $f(x|theta)$ be a gamily of PMFs/PDFs indexed by $\theta \in \Theta$. A family of distributions is $\Pi = \{\pi(\theta)\}$ is said to be conjugate for $f(x|\theta)$ if $\pi \in \Pi \implies \pi(\theta|x) \in \Pi$.

Note: Number of parameters in conjugate family is always going to be 1 more than the base distribution.

If $f(x|\theta)$ is an eponential family, $f(x|\theta) = (\pi h(x_i))e^{\sum\limits_{j=1}^{k} T_j(x)w_j(\theta)+nln[c(\theta)]}, T_j(x) = \sum\limits_{i=1}^{n} t_j(x_i)$

A conjugate family with hyperparameter $t \in \mathbb{R}^{k+1}$ is $\pi_t(\theta) \propto e^{\sum\limits_{j=1}^{k} t_j(x)w_j(\theta)+t_{k+1}ln[c(\theta)]}$

where t must satisfy $\int\limits_{\theta} e^{\sum\limits_{j=1}^{k} t_j(x)w_j(\theta)+t_{k+1}ln[c(\theta)]} < \infty$

Example: $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ is known

$$\begin{aligned} f(x|\theta) &= (2\pi\sigma^2)^{-n/2} e^{1/(2\sigma^2)\sum(x_i-\theta)^2} \\ &= h(x)e^{\frac{n\bar{x}}{\sigma^2}\theta - \frac{n}{2\sigma^2}\theta^2} \end{aligned}$$

Thus the conjugate prior has the form $\pi_t(\theta) \propto e^{t_1\theta+t_2\theta^2}$. This is e to a quadratic which must be a Normal. We know t_2 must be negative so the tails die out.

Since this must be a normal distirbution, find the mean and variance in terms of t_1 and t_2:

Let ν, τ^2 represent mean and variance that correspond to (t_1, t_2). $t_2 = \frac{-1}{2\tau^2}, t_1 = \frac{\nu}{\tau^2}$

$\pi_t(\theta) \propto e^{\frac{n\bar{x}}{\sigma^2}\theta - \frac{n}{2\sigma^2}\theta^2} e^{t_1\theta+t_2\theta^2} = e^{(t_1+\frac{n\bar{x}}{\sigma^2})\theta+(t_2-\frac{n}{2\sigma^2})\theta^2} = e^{t_1^*\theta+t_2^*\theta^2}$

$\implies \theta|x \sim N(\nu^*, \tau^{*2})$ where $\tau^{*2} = -\frac{1}{2t_2^*} = \frac{\sigma^2\tau^2}{n\tau^2+\sigma^2}, \nu^* = \frac{\tau^2}{\tau^2+\sigma^2/n}\bar{X} + \frac{\sigma^2/n}{\tau^2+\sigma^2/n}\nu$

3.3 Bayes Rules

Definition: A Bayes rule for a loss function $l(t|\theta)$ and a prior π is an estimator W^* for which the Bayes risk is minimized: $r_\pi(W^*) \leq r_\pi(W) \forall W$

$$
\begin{aligned}
r_\pi(W) &= \int_\Theta R(\theta; W)\pi(\theta)d\theta \\
&= \int_\Theta \mathbb{E}[l(W(x),\theta)|\theta]\pi(\theta)d\theta \\
&= \int_{\mathbb{X}} \mathbb{E}[l(W(x),\theta)|X=x]m(x)dx \\
&= \int_{\mathbb{X}} r_\pi(W(x)|x)m(x)dx
\end{aligned}
$$

If $r_\pi(W^*(x)|x) \leq r_\pi(W(x)|x)$, then W^* is the Bayes Rule

Example: $l(t,\theta) = (t-\theta)^2$

$$
\begin{aligned}
r_\pi(w(x)|x) &= \mathbb{E}[(w(x)-\theta)^2|X=x] \\
&= \int (w(x)-\theta)^2\pi(\theta|x)d\theta \\
&= \int_\Theta (w^2(x) - 2W(x)\theta + \theta^2)\pi(\theta|x)d\theta \\
&= w^2(x) - 2w(x)\,\mathbb{E}[\theta|X=x] + \mathbb{E}[\theta^2|X=x] \implies w^*(x) \\
&= \frac{2\,\mathbb{E}[\theta|X=x]}{2(1)} \\
&= \mathbb{E}[\theta|X=x]
\end{aligned}
$$

When using squared-error loss, the Bayes Rule is the posterior mean, when using absolute value squared-error loss it's the posterior median.

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REVIEW

$r_\pi(w) = \int_{\mathbb{X}} r_\pi(W(x)|x)m(x)dx$

$r_\pi(w(x)|x) = \mathbb{E}[l(W(x),\theta)|X=x]$

If $l(t,\theta) = (t-\theta)^2 \implies$ The Bayes Rule is $\mathbb{E}[\theta|X=x]$

If $l(t,\theta) = |t-\theta| \implies w^*(x)$ is the posterior median.

A reasonable loss smight be realted to SEL or AEL (Squared/Absolute Error Loss): $l(t,\theta) = g(\theta)(t-\theta)^2$

$$
\begin{aligned}
r_\pi(w(x)|x) &= \int\limits_\Theta l(t,\theta)\pi(\theta|x)d\theta \\
&= \int\limits_\Theta g(\theta)(t-\theta)^2\pi(\theta|x)d\theta \\
&= \int\limits_\Theta (t-\theta)^2[g(\theta)\pi(\theta|x)]d\theta
\end{aligned}
$$

Provided $\int\limits_\Theta [g(\theta)\pi(\theta|x)]d\theta$ is finite (integrable), it becomes the Kernel of another function: define: $\tilde{\pi}(\theta|x) \propto g(\theta)\pi(\theta|x)$

Then $r_\pi(w(x)|x) \propto \int\limits_\Theta (t-\theta)^2\tilde{\pi}(\theta|x)d\theta = r_{\tilde{\pi}}(w(x)|x)$ (under SEL) $\implies w^*(x) = \mathbb{E}_{\tilde{\pi}}(\theta|X=x)$

Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta)$

$\pi(\theta)$ is the $Gamma(\alpha, \beta)$

$l(t,\theta) = (\frac{t}{\theta} - 1)^2 = \frac{(t-\theta)^2}{\theta^2} \implies g(\theta) = \frac{1}{\theta^2}$

$g(\theta)\pi(\theta|x) \propto \frac{1}{\theta^2}\theta^n e^{\theta\sum x_i}\theta^{\alpha-1}e^{-\theta/\beta} = \theta^{\alpha+n-3}e^{-\theta(\sum x_i+\beta^{-1})}$ which is a $Gamma(\alpha+n-2, (\sum x_i + \beta^{-1})^{-1})$ (note that we need n ¿= 2 to have valid parameters).

The Bayes Rule is $w^*(x) = \frac{\alpha+n-2}{\sum x_i + \beta^{-1}}$

5 Best Unbiased Estimation

Recall: $MSE_\theta(w) = \mathbb{E}[(w-\theta)^2]$ (this is a risk function, so common it has its own name!)

Example: $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

1. $w(x) = \bar{x}, MSE_{\mu,\sigma^2}(\bar{x}) = \mathbb{E}_{\mu,\sigma^2}[(\bar{x}-\mu)^2]$

Since $\bar{x} \sim N(\mu, \sigma^2/n), MSE_{\mu,\sigma^2}(\bar{x}) = Var(\bar{x}) = \frac{\sigma^2}{n}$

2. Let $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

$MSE_{\mu,\sigma^2}(s^2) = \mathbb{E}_{\mu,\sigma^2}[(s^2-\sigma^2)^2] = Var(s^2) = \frac{\sigma^4}{(n-1)^2}2(n-1) = \frac{2\sigma^4}{n-1}$

In general:

$$
\begin{aligned}
MSE_\theta(w) &= \mathbb{E}_\theta[(w-\theta)^2] \\
&= \mathbb{E}_\theta[(w - \mathbb{E}_\theta[w] + \mathbb{E}_\theta[w] - \theta)^2] \\
&= \mathbb{E}_\theta[(w-\mathbb{E}_\theta[w])^2] + (\mathbb{E}_\theta[w]-\theta)^2 + 2\,\mathbb{E}_\theta[(w-\mathbb{E}_\theta[w])(\mathbb{E}_\theta[w]-\theta)] \\
&= Var_\theta(w) + Bias^2_\theta(w), Bias_\theta(w) = \mathbb{E}_\theta[w] - \theta
\end{aligned}
$$

Example: $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

$s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$

$\hat{\theta} = \frac{1}{n}\sum(x_i - \bar{x})^2$ (MLE/MM)

$$
\begin{aligned}
MSE_{\sigma^2}(\hat{\sigma}^2) &= Var_{\hat{\sigma}^2}(\hat{\sigma}^2) + Bias^2_{\sigma^2}(\hat{\sigma}^2) \\
&= Var_{\sigma^2}(\frac{n-1}{n}s^2) + Bias^2_{\sigma^2}(\frac{n-1}{n}s^2) \\
&= (\frac{n-1}{n})^2 \frac{2\sigma^4}{n-1} + [\mathbb{E}_{\sigma^2}(\frac{n-1}{n}s^2) - \sigma^2]^2 \\
&= \frac{2\sigma^4(n-1)}{n^2} + [\frac{n-1}{n}\sigma^2 - \sigma^2]^2 \\
&= \frac{2\sigma^4}{n-1}(\frac{n-1}{n})^2 + \frac{\sigma^2}{n^2} \\
&= \frac{2\sigma^4}{n-1}[\frac{(n-1/2)(n-1)}{n^2}] < MSE_{\sigma^2}(s^2)
\end{aligned}
$$

Let $\tau(\theta)$ be some estimand:

Define: w is said to be unbiased for $\tau(\theta)$ if $\mathbb{E}_\theta(w) = \tau(\theta) \forall \theta \in \Theta$

Definition: An estimator w^* is said to be a (uniformly) minimum variance unbiased estimator (MVUE) of $\tau(\theta)$ if:

1. w^* is unbiased for $\tau(\theta)$

2. For any w unbiased for $\tau(\theta), Var(w) \geq Var_\theta(w^*)$

Example: $x_1, ..., x_n \overset{\text{iid}}{\sim}, \mathbb{E}[x_i] = \theta$

Let $w_1 = x_1, w_2 = \bar{x}$. Both are unbiased, but $\bar{x}$ has smaller variance. To prove the MVUE, we'd have to compare w_2 to all other ws

Example: $x_i \overset{\text{iid}}{\sim} Pois(\lambda)$

Both $\bar{x}, s^2$ are unbiased. $\bar{x}$ is simpler, and also, $Var(\bar{x}) = \frac{\lambda}{n} < Var_\lambda(s^2)$

Two ways to establish that an estimator is an MVUE:

1. Find a lower bound $L(\theta)$ such that $Var_\theta(w) \geq L(\theta)$ for all unbiased w. Then, $Var_\theta(w^*) = L(\theta), w^*$ is MVUE

2. Show that the MVUE has characterstic properties and construct w^* with those properties

5.1 Cramer-Rao Lower Bound

The score function is $\psi(x;\theta) = \frac{\partial}{\partial\theta} ln(f(x|\theta))$

CR Condition: For any statistic W with finite expectation -

$(|\mathbb{E}_\theta[W]| < \infty \forall \theta)$, $\frac{d}{d\theta}\mathbb{E}_\theta[W] = \frac{d}{d\theta}\int\limits_{\mathbb{X}} W(x)f(x|\theta)dx = \int\limits_{\mathbb{X}} \frac{\partial}{\partial\theta} W(x)f(x|\theta)dx$

Notice: $\frac{\partial f(x|\theta)}{\partial\theta} = \frac{\partial f(x|\theta)}{\partial\theta}\frac{f(x|\theta)}{f(x|\theta)} = \frac{\partial ln(f(x|\theta))}{\partial\theta} f(x|\theta) = \psi(x;\theta)f(x|\theta)$

Thus: $\underline{\frac{d}{d\theta}\mathbb{E}_\theta[W] = \int\limits_{\mathbb{X}} W(x)\psi(x;\theta)f(x|\theta)dx}$

Definition: The Information Number (Fisher Information) is:

$I_n(\theta) = \mathbb{E}_\theta[\psi^2(x;\theta)]$ (Scalar)

$I_n(\theta) = \mathbb{E}_\theta[\psi(x;\theta)\psi(x;\theta)^\top]$ (Vector)

<u>Fact</u>: If CR condition holds, then the Fisher Information is the variance of the score:

$I_n(\theta) = Var_\theta(\psi(x;\theta))$ (note: this means that $\mathbb{E}_\theta[\psi(x;\theta)] = 0$)

<u>Theorem</u>: CRLB

Suppose the CR Condition holds and $0 < I_n(\theta) < \infty$. If W is a statistic satisfying:

i) $\mathbb{E}_\theta[W] = \tau(\theta), |\tau(\theta)| < \infty$ (unbiased estimator for $\tau(\theta)$) and

ii) $Var_\theta(W) < \infty \forall\theta$

then $Var_\theta(W) \geq \frac{(\tau'(\theta))^2}{I_n(\theta)}$ (for scalar θ)

Remember: Cauchy-Schwartz: $|x^\top y| \leq \|x\| \, \|y\| \implies |\,\mathbb{E}[xy] \leq \mathbb{E}[x^2]^{1/2}\,\mathbb{E}[y^2]^{1/2}$. Equality only happens when $x = ay$

<u>Proof</u>: CRLB

Apply CS inequality to $Y_\theta = W - \tau(\theta), z_\theta = \psi(x;\theta)$

$|\tau'(\theta)| = |\frac{d}{d\theta}\,\mathbb{E}_\theta[W]| = |\int\limits_{\mathbb{X}} W(x)\psi(x;\theta)f(x|\theta)dx| = |\,\mathbb{E}_\theta\, W\psi(x;\theta)|$ (here, because ψ has mean zero we can add constant on W)
$= |\,\mathbb{E}_\theta[Y_\theta z_\theta]| \leq [Var_\theta(Y_\theta)Var_\theta(z_\theta)]^{1/2} = [Var_\theta(W)I_n(\theta)]^{1/2} \implies Var_\theta(W) \geq \frac{(\tau'(\theta))^2}{I_n(\theta)}$

<u>Comments</u>:

1) To apply this result, recall that we have equality in the CRLB iff $W(X) - \tau(\theta) = k(\theta)\psi(x;\theta)$ (some constant times the score, refer to Cauchy Schwartz again). **THIS IS VERY IMPORTANT!!!!!**

2) In the iid case, let $I_1(\theta) = Var_\theta(\frac{d}{d\theta}ln(f(x_1|\theta)))$. Then,

$$
\begin{aligned}
I_n(\theta) &= Var_\theta(\frac{d}{d\theta}ln(f(\underset{\sim}{\boldsymbol{x}}|\theta))) \\
&= Var_\theta(\sum_{i=1}^n \frac{\partial}{\partial\theta}ln(f(\underset{\sim}{\boldsymbol{x}}|theta))) \\
&= \sum_{i=1}^n Var_\theta(\frac{\partial}{\partial\theta}ln(f(x_i|\theta))) \\
&= nI_1(\theta)
\end{aligned}
$$

3) If $\frac{d}{d\theta}\,\mathbb{E}_\theta[\psi(x;\theta)] = \int\limits_{\mathbb{X}} \frac{\partial}{\partial\theta}[\psi(x;\theta)f(x|\theta)]dx$ then: $I_n(\theta) = -\,\mathbb{E}_\theta[\frac{\partial}{\partial\theta}\psi(x;\theta)]$ (when iid:) $= -n\,\mathbb{E}_\theta[\frac{\partial^2}{\partial\theta^2}ln(f(x_1|\theta))]$

4) If $\tau(\theta)$ is a vector, $Var_\theta(W) \succeq \tau'(\theta)^\top I_n^{-1}(\theta)\tau'(\theta)$ ($A \succeq B$ means $A - B$ is Positive Definite)

<u>Example</u>: $X_i \overset{\text{iid}}{\sim} Pois(\theta), i = 1, ..., n$

a) Consider $\tau(\theta) = \theta$.

$$\begin{aligned}
\psi(x;\theta) &= \frac{\partial}{\partial\theta} ln[e^{-n\theta}\frac{\theta^{\sum x_i}}{\prod x_i!}] \\
&= \frac{\partial}{\partial\theta}[-n\theta + ln(\theta)\sum x_i - \sum ln(x_i!)] \\
&= -n + \frac{\sum x_i}{\theta} \\
&= \frac{n}{\theta}(\bar{x} - \theta) \\
&\implies \frac{\theta}{n}\psi(x;\theta) \\
&= \bar{x} - \theta \implies \bar{x} \text{ is the MVUE of } \theta.
\end{aligned}$$

To find $Var_\theta(\bar(x))$, we know that $Var_\theta(\bar(x)) = \frac{\theta}{n}$. We can also use the CRLB since it's an equality: $\tau'(\theta) = 1, I_n(\theta) = nI_1(\theta) = n(-\mathbb{E}_\theta[\frac{\partial}{\partial\theta}ln(f(x_1|\theta))]) = -n\,\mathbb{E}_\theta[\frac{\partial}{\partial\theta}(-1 + \frac{x_1}{\theta})] = n\,\mathbb{E}_\theta[\frac{x_1}{\theta^2}] = \frac{n\theta}{\theta^2} = \frac{n}{\theta} \implies CRLB = \frac{1}{n/\theta} = \frac{\theta}{n}$

b) Consider $\tau(\theta) = \theta^2$

Done in class on paper: $\bar{x}^2$ is a biased estimator: $\mathbb{E}[\bar{x}^2] = Var(\bar{x}) + \mathbb{E}[\bar{x}]^2 = Var(\bar{x}) + \theta^2$. The bias is $Var(\bar{x}) = \theta/n$

Example: $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \theta = (\mu, \sigma^2)$

$$I_n(\theta) = \begin{bmatrix} n/\sigma^2 & 0 \\ 0 & n/(2\sigma^4) \end{bmatrix}$$

$\tau(\theta) = \mu, \tau'(\theta) = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$

If W is unbiased for μ, $Var_\theta(W) \geq [\tau'(\theta)]^\top I_n^{-1}(\theta)\tau'(\theta) = \left(\begin{smallmatrix}1 & 0\end{smallmatrix}\right)\begin{pmatrix}\sigma^2/n & 0 \\ 0 & 2\sigma^4/n\end{pmatrix}\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) = \sigma^2/n \implies \bar{x}$ is MVUE.

$\tau(\theta) = \sigma^2 \implies \tau'(\theta) = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$

If W is unbiased for σ^2: $Var_\theta(W) \geq 2\sigma^2/n$

$Var_\theta(s^2) = 2\sigma^4/(n-1) > 2\sigma^4/n$

Example: (CR Condition Fails)

$X_i \overset{\text{iid}}{\sim} Unif(0,\theta)($ implicitly assuming $\tau(theta) = \theta)$

$\psi(x;\theta) = \frac{\partial}{\partial\theta}ln[f(x|\theta)] = \frac{\partial}{\partial\theta}ln[\theta^{-n}I(0 < X_{(n)} \leq \theta)]$

We can still compute $\mathbb{E}_\theta[\psi^2(x;\theta)] = \mathbb{E}_\theta[(\frac{\partial}{\partial\theta}(-nln(\theta)))^2] = \frac{n^2}{\theta^2} \implies$ The CRLB would be: $\frac{1}{n^2/\theta^2} = \frac{\theta^2}{n^2}$

$W(x) = \frac{n+1}{n}X_{(n)}$ is unbiased for θ. However, $Var_\theta(W) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{n^2}$. This is possible because the CR Condition didn't hold in the first place, so the CRLB wasn't actually a valid representation of the lower bound.

6 Constructing MVUEs using Completeness and Sufficiency

Let W be an unbiased estimator of $\tau(\theta)$ and T be any statistic. Consider $\mathbb{E}_\theta[W|T]$ as a new random variable.

$\mathbb{E}_\theta[\mathbb{E}_\theta(W|T)] = \mathbb{E}_\theta[W]$

If I condition in a way that I still get a statistic, it remains an unbaised estimator.

$Var_\theta(W) = Var_\theta(\mathbb{E}_\theta[W|T]) + \mathbb{E}_\theta[Var_\theta(W|T)] \geq Var_\theta(\mathbb{E}_\theta[W|T])$

Questions moving forward:

1. How can we choose T so that $\mathbb{E}_\theta[W|T]$ doesn't depend on θ? (sufficiency)

2. How can we choose T to maximally reduce the variance? (completeness)

6.1 Sufficiency

Because W is a statistic, if $x|T$ does not depend on θ, then neither will W.

Definition: A statistic T is sufficient if the distribution of $X|T$ does not depend on θ.

Interpretation: If T is sufficient, then two data points x, y should lead to the same inference if $T(x) = T(y)$.

Obvious Sufficient Statistics include:

- $x = (x_1, ..., x_n)$

- $(x_{(1)}, ..., x_{(n)})$

Fact: If I know $T(x)$ I can generate a new dataset Y with the exact same distributionas X and $T(x) = T(y)$.

Example: $x_1, ..., x_n \sim Bern(\theta)$. Then $\sum x_i$ is sufficient.

Generate Y using $\sum x_i$ such that $Y_i \overset{\text{iid}}{\sim} Bern(\theta)$

$P(Y = y|\sum x_i = t) = \begin{cases} 0 & \sum y_i \neq t \\ \frac{1}{\binom{n}{t}} & \sum y_i = t \end{cases}$

$P_\theta(Y = y) = \sum_{t=0}^{n} \frac{1}{\binom{n}{t}} I(\sum y_i = t) P_\theta(\sum x_i = t) = \sum_{t=0}^{n} I(\sum y_i = t)\binom{n}{t}\theta^t(1-\theta)^{n-t} = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$

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Sufficiency: If we take an unbiased estimator, and condition it with a sufficient statistic, it will stay an unbiased estimator that does not depend on θ.

If T is sufficient (discrete case), $P_\theta(X = x|T(X) = t) = \frac{P_\theta(X=x \text{ and } T(X)=t)}{P_\theta(T(x)=t)} = \frac{P_\theta(X=x)I(t=T(x))}{P_\theta(T(X)=T(x))} = \begin{cases} 0 & t \neq T(x) \\ \frac{p(x|\theta)}{q(T(x)|\theta)} & t = T(x) \end{cases}$

Theorem: Factorization

A statistic $T(X)$ is sufficient iff there are functions $h(X)$ and $g(t, \theta)$ such that $f(X|\theta) = h(x)g(T(x), \theta)$.

Proof (Discrete):

If T is sufficient, take $g(t, \theta) = q(t|\theta), h(x) = \frac{p(x|\theta)}{q(T(x)|\theta)}$. That yields the same as above.

Now, assume the factorization holds: $q(t|\theta) = P_\theta(T(X) = t) = \sum_{y:T(y)=t} p(y|\theta) = \sum_{y:T(y)=t} h(y)g(T(y), \theta) = [\sum_{y:T(y)=t} h(y)]g(t, \theta)$.

Now, $\frac{p(x|\theta)}{q(T(x)|\theta)} = \frac{h(x)g(T(x),\theta)}{[\sum\limits_{y:T(y)=T(x)} h(y)]g(T(x),\theta)} = \frac{h(x)}{\sum\limits_{y:T(y)=T(x)} h(y)}$ which does not involve θ.

Example:

$X_i \overset{\text{iid}}{\sim} N(\theta, 1)$

$f(x|\theta) = \prod\limits_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x_i-\theta)^2} = (2\pi)^{-n/2} exp(-1/2 \sum\limits_{i=1}^{n}(x_i-\theta)^2)$

Example factorizations

1. $(2\pi)^(-n/2)$: $h(x)$ or could be absorbed into g (doesn't really matter)

$g(T(x),\theta)$ is the exponential: $T(x) = (x_1, ..., x_n)$

2. $(2\pi)^{-n/2} exp(-1/2 \sum\limits_{i=1}^{n}(x_{(i)}-\theta)^2)$

$T(x) = (x_{(1)}, ..., x_{(n)})$

3. $(2\pi)^{-n/2} exp(-1/2 \sum x_i^2) exp(n\bar{x}\theta - \frac{n\theta^2}{2})$

$\implies T(X) = \sum x_i$ or $T(X) = \bar{x}$ are sufficient

Example: $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \theta = (\mu, \sigma^2)$

$f(x|\theta) = (2\pi\sigma^2)^{-n/2} exp(-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2) = (2\pi)^{-n/2}\sigma^{-n} exp(-\frac{1}{2\sigma^2}\sum(x_i^2) + \frac{\mu}{\sigma^2}\sum x_i - \frac{n\mu^2}{2\sigma^2})$

$T(x) = (\sum x_i, \sum x_i^2)$ is sufficient.

Note: Any 1:1 transformation of a sufficient statistic is sufficient!

Set $t_1 = \sum x_i, t_2 = \sum x_i^2$: $\bar{x} = \frac{t_1}{n}, s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2 = \frac{1}{n-1}(\sum x_i^2 - n\bar{x}^2) = \frac{1}{n-1}(t_2 - \frac{t_1^2}{n}) \implies (\bar{x}, s^2)$ is also sufficient.

Example: $X_i \overset{\text{iid}}{\sim} Unif(\theta, \theta+1)$

$f(x|\theta) = \prod\limits_{i=1}^{n} I(x_i > \theta) I(x_i < \theta + 1) = I(X_{(1)} > \theta) I(X_{(n)} < \theta) \implies T(X) = (X_{(1)}, X_(n))$ is sufficient.

Example (Exponential Families)

$f(x|\theta) = (\prod h(x_i)) exp[\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{k} w_i(\theta) t_j(x_i) + nln(c(\theta))] = \tilde{h}(x) exp[\sum\limits_{j=1}^{k} w_j(\theta) T_j(x) + nln(c(\theta))]$

$\implies (T_1(x), ..., T_k(x))$ is a sufficient statistic.

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Definition: A sufficient statistic is is said to be minimal sufficient if it is a function of any other sufficient statistic.

Note: a function $g(x)$ is a function of $f(x)$ iff $f(x) = f(y) \implies g(x) = g(y)$

Let $T(X)$ be any sufficient statistic. Then by the factorization theorem, $f(x|\theta) = h(x)g(T(x),\theta)$. Therefore, when $T(x) = T(y)$, $\frac{f(x|\theta)}{f(y|\theta)} = \frac{h(x)g(T(x),\theta)}{h(y)g(T(y),\theta)} = \frac{h(x)}{h(y)}$: doesn't depend on θ. When minimally sufficient, this implies the ratio will not have a dependence on θ. We also want this to go the other way.

Theorem: A sufficient statistic is minimal sufficient if $\frac{f(x|\theta)}{f(y|\theta)}$ is independent of θ implies $T(x) = T(y)$.

Proof: Define T^* to be any sufficient statistic, and T satisfies the above theorem condition. If $T^*(x) = T^*(y)$ then by factorization theorem, $\frac{f(x|\theta)}{f(y|\theta)} = \frac{h^*(x)g^*(T^*(x),\theta)}{h^*(y)g^*(T^*(y),\theta)} = \frac{h^*(x)}{h^*(y)}$ which doesn't depend on θ.

By the above theorem, $T(x) = T(y)$, so $T(x)$ is minimal sufficient.

Example: $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

1) σ^2 known: $f(x|\mu) = (2\pi\sigma^2)^{-n/2} exp(-\frac{1}{2\sigma^2}\sum(x_i - \mu)^2) = (2\pi\sigma^2)^{n/2} exp(1\frac{1}{2\sigma^2} + \frac{n\mu\bar{x}}{\sigma^2} - \frac{n\mu^2}{2\sigma^2})$

$\frac{f(x|\mu)}{f(y|\mu)} = exp(-\frac{1}{\sigma^2}[\sum x_i^2 - \sum y_i^2])exp(\frac{n\mu}{\sigma^2}(\bar{x} - \bar{y}))$. The only way to have independence from μ is if $\bar{x} = \bar{y}$, so $T(x) = \bar(x)$ is minimal sufficient.

Example: $X_i \overset{\text{iid}}{\sim} Unif(0, 1)$

$f(x|\theta) = I(\theta < x_{(1)})I(\theta + 1 > x_{(n)})$

$\frac{f(x|\theta)}{f(y|\theta)} = \frac{I(\theta < x_{(1)})I(\theta+1 > x_{(n)})}{I(\theta < y_{(1)})I(\theta+1 > y_{(n)})}$. For this to be independent of θ, the corresponding indicators must 'change' at the same values, meaning $x_{(1)} = y_{(1)}$ and $x_{(n)} = y_{(n)}$. Therefore, $T = (x_{(1)}, x_{(n)})$ is minimal sufficient.

In this family, $T'(x) = (\frac{x_{(n)} - 1 + x_{(1)}}{2}, x_{(n)} - x_{(1)})$ is minimal sufficient. Notice $X_{(n)} - X_{(1)} = (X_{(n)} - \theta) - (X_{(1)} - \theta) \overset{D}{=} z_{(n)} - z_{(1)}, Z_i \overset{\text{iid}}{\sim} Unif(0, 1)$ meaning that part of the minimal statistic does not depend on θ: but the minimal statistic is supposed to be the minimal information to describe θ: uh oh

Definition: A statistic S is ancillary to θ if its distribution does not depend on θ.

Finding Ancillary Statistics:

1. Location Families: $f(x|\theta) = g(x - \theta)$. $X_i - \theta \overset{D}{=} Z \sim g \implies X_i - X_j$ is ancillary. Or $X_{(i)} - X_{(j)}$, etc. Spacing between the data is anciallry in a location family

2. Scale Families: $f(x|\theta) = \frac{1}{\theta}g(x/\theta), \theta > 0$. $\frac{X_i}{\theta} \overset{D}{=} Z \sim g \implies, \frac{X_i}{X_j}, \frac{X_{(i)}}{X_{(j)}}$.

9 Completeness

Want statistics that are "completely about θ"

Definition: A statistic T is complete for θ if, for a transformation $g(T), \mathbb{E}_\theta[g(T)] = 0 \implies g(T) = 0$ with probability 1.

Example: $X_i \overset{\text{iid}}{\sim} f(x_i|\theta)$

$T(X) = X$ is not complete. Let $g(T) = x_1 - x_2$. Then $P(g(T) \neq 0) > 0$, but $\mathbb{E}_\theta[g(T)] = \mathbb{E}_\theta[x_1] - \mathbb{E}_\theta[x_2] = 0$.

Example: $X_i \overset{\text{iid}}{\sim} Unif(\theta, \theta + 1)$

$T = (x_{(1)}, x_{(n)})$ is not complete. Let $g(T) = x_{(n)} - x_{(1)} - \mathbb{E}[X_{(n)} - x_{(1)}]$. Since $P(g(T) \neq 0) > 0, \mathbb{E}_\theta(g(T)) = 0$, T is not complete.

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Example: $X_i \stackrel{\text{iid}}{\sim} Unif(0,\theta), \theta > 0$

$T = X_{(n)}$ is minimal sufficient. Is it complete?

To show completeness, assume there exists a function $g(t)$ such that $\mathbb{E}_\theta[g(T)] = 0 \forall \theta > 0$. We then need to show that this only happens when $g(t) = 0$.

Assume $g(t)$ is continuous. Assume $0 = \mathbb{E}_\theta[g(T)] = \int\limits_0^\theta g(t)X_{(n)}dt = \int\limits_0^\theta g(t)n\frac{t^{n-1}}{\theta^n}dt$

Derivative of both sides:

$$0 = n\frac{d}{d\theta}\theta^{-n}\int\limits_0^\theta g(t)t^{n-1}dt = n\left[-n\theta^{-n+1}\int\limits_0^\theta g(t)t^{n-1}dt + \theta^{-n}g(\theta)\theta^{n-1}\right] = -n^2\theta^{-1}\int\limits_0^\theta g(t)\frac{t^{n-1}}{\theta^n}dt + \frac{ng(\theta)}{\theta} = \frac{ng(\theta)}{\theta}$$

$\implies 0 = \frac{ng(\theta)}{\theta} \implies 0 = g(\theta) \forall \theta > 0$, so $T = X_{(n)}$ is complete.

Example: $X_i \stackrel{\text{iid}}{\sim} Bern(\theta), 0 < \theta < 1$

$T = \sum X_i$ is sufficient (+ minimal).

Assume $g(t)$ satisfies $\mathbb{E}_\theta[g(T)] = 0$

$$0 = \mathbb{E}_\theta[g(T)] = \sum_{i=1}^n g(t)\binom{n}{t}\theta^t(1-\theta)^{n-t} = (1-\theta)^n\sum_{i=1}^n g(t)\binom{n}{t}\left(\frac{\theta}{1-\theta}\right)^t \forall 0 < \theta < 1$$

This is a polynomial of degree n in $r = \frac{\theta}{1-\theta} \in (0,\infty)$

$\implies$ the polynomial must be 0, meaning all of its coefficients must be 0.

$\implies g(t)\binom{n}{t} = 0 \implies g(t) = 0, t = 0, ..., n$

$\implies T$ is complete.

Theorem:

Assume $X_i \stackrel{\text{iid}}{\sim} f(x|\theta)$ where $f(x|\theta)$ is an exponential family:

$$F(x|\theta) = h(x)exp\{\sum_{j=1}^k w_j(\theta)t_j(x) + ln(c(\theta))\}$$

The sufficient statistic $T = (\sum_{i=1}^n t_1(x_i), ..., \sum_{i=1}^n t_k(x_i))$ is also complete iff the parameter space $\Theta \subset \mathbb{R}^k$ contains an open set.

Interval $(0,1)$ is open. All intergers is not open. Open: can you 'draw' an infitessmally small 'circle' around some point in your set and still only contain things in your set?

Example: $X_i \stackrel{\text{iid}}{\sim} N(\theta_1,\theta_2), T = (\bar{X}, S^2)$

i) $\Theta = \mathbb{R}x(0,\infty)$ is open. Then T is complete

ii) $\Theta = \mathbb{R}x\{1,2\}$ is not open. The second set is not open, so T is not complete.

iii) $\Theta = \{(\theta_1,\theta_2) : \theta_2 = \theta_1^2 + 1\}$ which is not an open set, so T is not complete.

For (ii) and (iii), try drawing out the support and see if you can draw a circle around any point and still be in the set.

Facts: Facts about complete statistics:

1) If a minimal sufficient statistic exists, then any complete sufficient statistic is also minimal sufficient

2) Any complete sufficient statistic is independent of any ancillary statistic

Example: $X_i \overset{\text{iid}}{\sim} N(\theta, 1), \theta \in \mathbb{R}$

Then $T = \bar{X}$ is complete and sufficient.

Since $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2 = \frac{1}{n-1}\sum[(x_i - \theta) - (\bar{x} - \theta)]^2 \overset{D}{=} \frac{1}{n-1}\sum(z_i - \bar{z})^2, z_i \overset{\text{iid}}{\sim} N(0, 1)$

$\implies S^2$ is anciallary, $\implies \bar{X} \perp\!\!\!\perp S^2$

If instead $Y_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2 > 0$ now not known.

What about $(\bar{Y}, S^2_Y)$

$X_i oversetD{=}Z_i + \theta \implies \bar{X} \overset{D}{=} \bar{Z} + \theta$

$Y_i \overset{D}{=} \sigma Z_i + \theta \implies \bar{Y} \overset{D}{=} \sigma\bar{Z} + \theta \implies \bar{Y} \overset{D}{=} \sigma(\bar{X} - \theta) + \theta$

$S^2 \overset{D}{=} S^2_Z$

$S^2_Y \overset{D}{=} \sigma^2 S^2_Z \implies (\bar{Y}, S^2_Y) \overset{D}{=} (\sigma(\bar{X} - \theta) + \theta, \sigma^2 S^2) \implies$ these are independent.

10 Returning to MVUEs

Assume W is an unbiased estimator for $\tau(\theta)$ and that T is an sufficient statistic.

Theorem: Rao-Blackwell

The statistic $\phi(T) = \mathbb{E}[W|T]$ is unbiased for $\tau(\theta)$ and $Var_\theta(\phi(T)) \leq Var_\theta(W)$

Proof:

i) $\phi(T)$ is a statistic because T is sufficient

ii) $\phi(T)$ is unbiased because $\mathbb{E}_\theta[\phi(T)] = \mathbb{E}_\theta[\mathbb{E}(W|T)] = \mathbb{E}_\theta[W] = \tau(\theta)$

iii) $Var_\theta(W) = Var_\theta(\phi(T)) + \mathbb{E}_\theta[Var_\theta(W|T)] \geq Var_\theta(\phi(T))$

Theorem:

If an MVUE exists, then it is unique

Proof:

Let W_1, W_2 be MVUEs. Define $W_3 = \frac{1}{2}(W_1 + W_2)$ (unbiased), $Var_\theta(W_3) = \frac{1}{4}[Var_\theta(W_1) + Var_\theta(W_2) + 2Cov_\theta(W_1, W_2)] = \frac{1}{2}[Var_\theta(W_1) + 2Cov_\theta(W_1, W_2)] \leq Var_\theta(W_1)$

Must have equality on the last step (otherwise W_1, W_2 would not be MVUEs), implying $W_1 = W_2$.

Theorem:

An unbiased estimator W is the MVUE iff it is uncorrelated with any U satistfying $\mathbb{E}_\theta[U] = 0, \forall\theta$ (unbiased estimators of 0).

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Theorem:

If W is unbiased for $\tau(\theta)$, then W is the MVUE iff it is uncorrelated with any unbiased estimator of 0.

Proof:

Assume W is uncorrelated with any U satisfying $\mathbb{E}_\theta[U] = 0$. Then if W' is any unbiased estimator of $\tau(\theta)$, then $U = W - W'$ is unbiased for 0. $Var_\theta(W') = Var_\theta(W - U) = \text{Var}_\theta(W) + \text{Var}_\theta(U) - 2Cov_\theta(W, U) = \text{Var}_\theta(W) + \text{Var}_\theta(U) \geq \text{Var}_\theta(W)$

Hint for HW: Assume the $Cov_\theta(W, U) \neq 0$ for some unbiased U for 0. Then consider $W' = W + aU$. Then use contrapositive to show we can find something with lower variance.

Theorem: (Lehman-Scheffe)

Suppose W is a function of a complete sufficient statistic. Then it is the MVUE of $\tau(\theta) = \mathbb{E}_\theta(W)$.

Proof:

Let U be unbiased for 0. $Cov_\theta(W, U) = \mathbb{E}_\theta(WU) = \mathbb{E}_\theta[\mathbb{E}_\theta[WU|T]] = \mathbb{E}_\theta[W\,\mathbb{E}_\theta[U|T]]$. Because T is sufficient, $g(T) = \mathbb{E}_\theta(U|T)$ doe snot depend on θ. However, $\mathbb{E}_\theta[g(T)] = \mathbb{E}_\theta[U] = 0$. By completeness, $P(g(T) = 0) = 1$.

$\implies Cov_\theta(W, U) = \mathbb{E}_\theta[W(0)] = 0$

Example: $X_i \overset{\text{iid}}{\sim} Pois(\theta), \tau(\theta) = \theta^2, \theta > 0$

We showed previously that $W = \bar{X}^2 - \frac{\bar{X}}{n}$ is unbiased for θ^2.

Why is it the MVUE? Because $\bar{X}$ is complete (exponential family, open set) and sufficient (factorization theorem).

Example: $X_i \overset{\text{iid}}{\sim} N(\theta, 1), \theta \in \mathbb{R}$. Find MVUE for $\tau(\theta) = \theta^2$.

$\bar{X}$ was the MVUE for $\tau(\theta) = \theta$, so we'll start with a guess of $W = \bar{x}^2$. $\mathbb{E}_\theta[W] = \text{Var}_\theta(\bar{x}) + \mathbb{E}_\theta[\bar{x}]^2 = \frac{1}{n} + \theta^2$. So the bias is $\frac{1}{n}$.

So, $W = \bar{x}^2 - \frac{1}{n}$ is the MVUE.

Remember, for any distribution, $\mathbb{E}_\theta[\bar{x}] = \theta$ (population mean) and $\text{Var}_\theta(\bar{x}) = \frac{\sigma^2}{n}$.

Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta), \theta > 0$. $P(X_1 \leq 1) = \tau(\theta) = 1 - e^{-\frac{1}{\theta}}$.

1. What is the complete sufficient statistic?

$\theta^{-n} e^{-\frac{\sum X_i}{\theta}} \implies \sum X_i$ is CSS (Complete Sufficient Statistic).

2. Start with any unbiased estimator

$W = I(X_1 \leq 1)$ is unbiased, so $W^* = \mathbb{E}[W|T]$ is the MVUE.

$\frac{X_1}{\sum X_i} \sim Beta(1, n-1)$ is independent of T

$$\mathbb{E}[I(X_1 \leq 1)|T] = P(\tfrac{X_1}{T} \leq \tfrac{1}{T}|T) = \begin{cases} 1 & T < 1 \\ 1 - (1 - \frac{1}{T})^{n-1} & T \geq 1 \end{cases}$$

Summary:

Can use Cramer-Rao lower bound to find MVUE if you're able to factor the Score. Otherwise, dont get a whole lot.

Exponential Family: Sufficiency is automatic, completeness depends on support

Suffifiency: Factorization Theorem. Simplify correctly, end with Minimal Sufficient

Lehman-Scheffe: If something is Complete and Suffificient, it is the MVUE

12 Asymptotic Evaluation of Estimators

Looking at 'as n goes to infinity', or large sample theory.

Definition: Let W_n be an estimator based on sample of size n of $\tau(\theta)$. W_n is said to be consistent for $\tau(\theta)$ if $W_n \overset{P_\theta}{\to} \tau(\theta) \forall \theta$

$P_\theta(|W_n - \tau(\theta)| > \epsilon) \underset{n \to \infty}{\to} 0$ for all θ and any $\epsilon > 0$.

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Methods for showing consistency:

1. Tail/Concentration Bounds to control the probabilities

Markov's Inequality, Chebychev, Chernoff

2. If $W_n = g(\bar{x})$, we can apply LLN ($\bar{x} \overset{P}{\to} \mu$) then CMT (Continous Mapping Theorem) ($G(\bar{x} \overset{P}{\to} g(\mu)))$ if g is continous.

3. M-estimation (Maiximizer/Minimizer of some function, ie. MLE)

$\frac{1}{n} l(\theta|x) \overset{P_{\theta_0}}{\to} E_{\theta_0}[ln(f(x_1|\theta))]$

Basically, we know the right side is min/max at θ_0, so we need to show that the left side approcahes that same min/max as the overeall distirbutions converge

Example:

Suppose W_n satisfies:

1) $\mathbb{E}_\theta[W_n] \underset{n \to \infty}{\to} \tau(\theta)$ (W_n is asymptocially unbiased for $\tau(\theta)$)

2) $Var_\theta \underset{n \to \infty}{\to} 0$

In this case, W_n is consistent:

$P_\theta(|W_n - \tau(\theta)| > \epsilon) \leq \frac{\mathbb{E}_\theta[(W_n - \tau(\theta))^2]}{\epsilon^2}$ Chebychev's! $= \frac{MSE_{\tau(\theta)}(W_n)}{\epsilon^2} \to 0$ by (1) and (2).

Example: $X \overset{\text{iid}}{\sim} Beta(\alpha, 1)$

1) Method of Moments: $\bar{x} = \frac{\alpha}{\alpha+1} \leftrightarrow \hat{\alpha} = \frac{\bar{x}}{1-\bar{x}}$

$\bar{X} \overset{P}{\to} \frac{\alpha}{\alpha+1}$ as $n \to \infty$

Define $g(x) = \frac{x}{1-x}$, continious on (0, 1). By CMT: $g(\bar{x}) \overset{P}{\to} g(\frac{\alpha}{\alpha+1}) = \frac{\frac{\alpha}{\alpha+1}}{1-\frac{\alpha}{\alpha+1}} = \alpha$

Thus, MoM estimator converges to α so it is consistent.

2) The MLE is $\hat{\alpha} = \frac{n}{-\sum ln(x_i)} = \frac{-1}{\bar{Y}}, \bar{Y} = \frac{-1}{n}\sum ln(x_i)$

$\bar{Y} \xrightarrow{P} \mathbb{E}_\alpha[-ln(x_i)], -ln(x_i) \sim Exp(\frac{a}{\alpha}) \implies \bar{Y} \xrightarrow{P} \frac{1}{\alpha}$

$g(x) = \frac{1}{x}$ continous at $\frac{1}{\alpha}$, so $g(\bar{Y}) \xrightarrow{P} \alpha$

The CLT says as $\sqrt{n}(\bar{X} - \mu) \xrightarrow{D} N(0, \sigma^2)$

$P(|\bar{X} - \mu| > \frac{C}{\sqrt{n}}) = P(\sqrt{n}|\bar{X} - \mu| > C) \to 1 - \Phi(\frac{C}{\sigma})$

Example: $X \overset{\text{iid}}{\sim} Unif(0, \theta)$

$\frac{X_{(n)}}{\theta} \sim Beta(n, 1)$, so $P_\theta(|X_1 - \theta| > Cn^{-1}) = P_\theta(\theta - X_{(n)} < Cn^{-1}) = P_\theta(\frac{X_{(n)}}{\theta} < 1 - \frac{C}{n\theta}) = (1 - \frac{C}{n})^n \to e^{-C}$

Definition: If $\lim_{n\to\infty} b_n Var(W_n) \to \gamma^2$, then γ^2 is called the limiting variance of W_n

$Var(W_n) \approx \frac{\gamma^2}{n}$

eg. $Var(\bar{X}) = \frac{\sigma^2}{n} \implies b_n = n, b_n Var(\bar{X}) = \sigma^2 \to \sigma^2$

Example: $X \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

$Var(\frac{1}{\bar{X}}) = \infty$

Definition: Suppose $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{D} Y, E(Y) = 0$, and $Var(Y) < \infty$. Then, $\sigma^2 = Var(Y)$ is the asymptotic variance of W_n

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Definition (same as end of last class, but restated) If $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{D} Y_\theta, \forall\theta$, where $\mathbb{E}_\theta[Y_\theta] = 0$, then $\text{Var}_\theta(Y_\theta)$ is called the asymptotic variance of W_n.

$\text{Var}_\theta(W_n) = \frac{1}{n}\text{Var}_\theta(\sqrt{n}(W_n - \tau(\theta))) \approx \frac{\text{Var}_\theta(Y_\theta)}{n}$

Example: $X \overset{\text{iid}}{\sim} Pois(\lambda), \lambda > 0$

The MLE of $\tau(\lambda) = \lambda^2$ is $\bar{X}^2$, while the MVUE is $\bar{X}^2 - \frac{\bar{X}}{n}$

Notice: $\bar{X}^2 - (\bar{X}^2 - \frac{\bar{X}}{n}) = \frac{\bar{X}}{n} \xrightarrow{P} 0$ meaning the MLE and MVUE converge as n increases. Even if you multiply by $\sqrt{n}$ this still converges in probability to 0.

"1-2" punch for MVUEs: Law of Large numbers, continous mapping theorem

"1-2" punch for Asymptotics: Central Limit Theorem, Delta Method

In this case, CLT: $\sqrt{n}(\bar{X} - \lambda) \xrightarrow{D} N(0, \lambda)$ (Note, n cannot appear in the limiting distribution)

Delta Method: $\sqrt{n}(g(\bar{x}) - g(\lambda)) \xrightarrow{D} N(0, \lambda[g'(\lambda)]^2), g(x)$ in our case is x^2, $g'(x) = 2x$.

$\implies \sqrt{n}(\bar{X}^2 - \lambda^2) \xrightarrow{D} N(0, 4\lambda^3)$

Thus the asymptotic variance is $4\lambda^3$, and the asymptotic variance approximation is: $Var_\lambda(\bar{X}) \approx \frac{4\lambda^3}{n}$

The CRLB is: $\frac{(2\lambda)^2}{n/\lambda} = \frac{4\lambda^3}{n}$

Define: An estimator W_n is said to be:

i) Asymptotically unbiased if $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{D} Y_\theta, \mathbb{E}_\theta[Y_\theta] = 0$

ii) Asymptotically efficient if its asymptotic varaince is $\frac{[\tau'(\theta)]^2}{I_1(\theta)}$ (the CRLB, Fischer information calculated for 1 sample)

Theorem: Under regularity conditions, if $X \overset{\text{iid}}{\sim} f(x|\theta)$, then:

i) the MLE ($\hat{\theta}_n$) will exist and be unique with probability converging to 1

ii) $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, I_1^{-1}(\theta_0))$

Proof:

$l(\theta|x) = \sum_{i=1}^{n} ln(f(x_i|\theta)) \implies l'(\theta|x) = \psi(X;\theta) = l'(\theta_0|x) + l''(\theta_0|x)(\theta - \theta_0) + R_n(\theta, \theta_0)$

By regularity conditions, $\frac{R_n(\hat{\theta}_n, \theta_0)}{\sqrt{n}} \xrightarrow{P} 0$

Plug in $\theta = \hat{\theta}_n$, by MLE: $0 = l'(\theta_0|x) + l''(\theta_0|x)(\hat{\theta}_n - \theta_0) + R_n(\hat{\theta}_n, \theta_0)$

$\implies \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{\sqrt{n}l'(\theta_0|x)}{-l''(\theta_0|x)} + \frac{\sqrt{n}R_n(\hat{\theta}_n,\theta_0)}{-l''(\theta_0|x)}$

Look at $-l''(\theta_0|x) = -\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta^2} ln(f(X_i|\theta))|_{\theta=\theta_0} \implies \frac{1}{n}(-l''(\theta_0|x)) \xrightarrow{P_{\theta_0}} \mathbb{E}_{\theta_0}[-\frac{\partial^2}{\partial\theta^2} ln(f(X_i|\theta))|_{\theta=\theta_0}] = I_1(\theta_0)$

Now examine: $\frac{1}{\sqrt{n}} \sum_{i=}^{n} l'(\theta_0|x) = \frac{1}{\sqrt{n}}(\sum_{i=1}^{n} \psi(X_i;\theta_0)) = \sqrt{n}(\frac{1}{n}\sum_{i=1}^{n} \psi(X_i;\theta_0))$

We know $\mathbb{E}_{\theta_0}(\psi(X_i;\theta_0)) = 0$, $Var_{\theta_0}(\psi(X_i;,\theta_0)) = I_1(\theta_0)$

Apply CLT to $Y_i = \psi(X_i;\theta_0) : \sqrt{n}\bar{Y} = \frac{1}{\sqrt{n}} l'(\theta|x) \xrightarrow{D} N(0, I_1(\theta_0))$

By Slutsky's theorem (as n inreases, anything that converges to something with mean 0 dissappears):

$\frac{\sqrt{nl'(\theta_0|x)}}{-l''(\theta_0|x)} = \frac{\frac{1}{\sqrt{n}}l'(\theta_0|x)}{-\frac{1}{n}l''(\theta_0|x)} \xrightarrow{D} \frac{N(0,I_1(\theta_0))}{I_1(\theta_0)} = N(0, I_1^{-1}(\theta_0))$

$\frac{\sqrt{n}R_n(\hat{\theta}_n,\theta_0)}{-l''(theta_0|x)} = \frac{\frac{1}{\sqrt{n}}R_n(\hat{\theta}_n,\theta_0)}{-\frac{1}{n}l''(\theta_0|x)} \xrightarrow{P_{\theta_0}} 0$

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Example: $X \overset{\text{iid}}{\sim} Bern(\theta), 0 < \theta < 1$

The MLE is $\bar{X}_n$ as long as $\bar{X} \not\in \{0,1\}$

The CLT says $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{D} N(0, \theta(1-\theta))$

The MLE result says that $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{D} N(0, I_1^{-1}(\theta))$

Recall that $I_1(\theta) = -\mathbb{E}_\theta[\frac{\partial^2}{\partial\theta^2} ln(\theta^x(1-\theta)^{1-x})] = -\mathbb{E}_\theta[\frac{\partial^2}{\partial\theta^2}(Xln(\theta) + (1-X)ln(1-\theta))] = \mathbb{E}_\theta[\frac{X}{\theta^2} + \frac{1-X}{(1-\theta)^2}] = \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)}$

What about MLE for $\eta = \tau(\theta) = ln(\frac{\theta}{1-\theta})$

$\tau'(\theta) = \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)} \implies \sqrt{n}(\hat{\eta} - \eta) \xrightarrow{D} N(0, \frac{1}{\theta(1-\theta)})$

15.1 Relative Efficiency

Suppose W_n and V_n are both asymptocially unbiased and have asymptotic variances σ_W^2, σ_V^2

Definition: The relative asymptotic efficiency of V_n compared to W_n is $ARE(V_n, W_n) = \frac{\sigma_W^2}{\sigma_V^2}$ (an estimator is better when it's asymptotic variance is smaller, so this says that relative efficiency increases as the target estimator's variance decreases comapred to another's).

Example: $X \overset{\text{iid}}{\sim} Pois(\lambda), \lambda > 0, \tau(\lambda) = P_\lambda(X_1 = 0) = e^{-\lambda}$

Get some naive estimates for λ:

$W_n = \frac{1}{n}\sum_{i=1}^{n} I_{(X_J=0)}, V_n = e^{-\bar{X}}$

Now find the asymptotic variances:

We know that $\sqrt{n}(W_n - e^{-\lambda}) \xrightarrow{D} N(0, e^{-\lambda}(1-e^{-\lambda}))$. Also, $\sqrt{n}(V_n - e^{-\lambda}) \xrightarrow{D} N(0, e^{-\lambda}I_1(\theta)) = N(0, \lambda e^{-2\lambda})$

$\implies ARE(V_n, W_n) = \frac{e^{-\lambda}(1-e^{-\lambda})}{\lambda e^{-2\lambda}} = \frac{e^{\lambda}(1-e^{-\lambda})}{\lambda} = \frac{e^{\lambda}-1}{\lambda} > 1$

Example: Mean vs. Median: $X \overset{\text{iid}}{\sim} f$, f is symmetric about some point θ, and $\mathbb{E}[X_i] = \theta$

$\sqrt{n}(\bar{X} - \theta) \xrightarrow{D} N(0, \sigma^2), \sigma^2 = Var(x_i)$

$\sqrt{n}(\tilde{X} - \theta) \xrightarrow{D} N(0, \frac{1}{2f(\theta)^2})$

$ARE(\tilde{(X)}, \bar{X}) = 4\sigma^2 f^2(\theta)$

i) Normal: $ARE(\tilde{X}, \bar{X}) = 4\sigma^2(2\pi\sigma^2)^{-1} = \frac{2}{\pi} \approx 0.64$

ii) Laplace: $f(x|\beta, \theta) = \frac{1}{2\beta}e^{\frac{-|x-\theta|}{\beta}}$: $ARE(\tilde{X}, \bar{X}) = 4(2\beta^2)(\frac{1}{2\beta})^2 = 2$ (smaller tails, median is more efficient)

15.2 Hypothesis Testing

$X_1, ..., X_n \sim f(X|\theta)$

Definition: Let $\theta \in \Theta$. A null hypothesis is a statement $H_0 : \theta \in \Theta_0 \subset \Theta$. The corresponding alternative hyposthesis is: $H_1 : \theta \in \Theta_1 = \Theta \setminus \Theta_0$

Example: $X \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$

i) σ^2 known, $\theta = \mu, \Theta = \{0, 1\}$

$H_0 : \mu = 0 : [\theta \in \Theta_0 = \{0\}]$

$H_1 : \mu = 1 : [\theta \in \Theta_1 = \{1\}]$

ii) σ^2 known, $\theta = \mu$

$H_0 : \theta \in \Theta_0 = [0, \infty)$

$H_1 : \theta \in \Theta_1 = (-\infty, 0)$

iii) σ^2 unknown: $\theta = (\mu, \sigma^2)$

$H_0 : \mu = 0 : [\theta \in \Theta_0 = \{0\} \times (0, \infty)]$

$H_1 : \mu \neq 0 : [\theta \in \Theta_1 = (R \setminus \{0\}) \times (0, \infty)]$

Definition: Let $\mathbb{X}$ be the sample space. A test funciton is a statistic $\phi(x) \in [0, 1]$ such that if $x \in \mathbb{X}$, we reject H_0 with probability $\phi(x)$.

Example: $X \overset{\text{iid}}{\sim} Bin(n, \theta)$

Some test functions:

$\phi_1(x) = I(X > 5)$

$\phi_2(x) = \frac{X}{n}$

$$\phi_3(x) = \begin{cases} 0, & x < k \\ \gamma, & x = k \\ 1, & x > k \end{cases}$$

Definition: A simple test function satisfies $\phi(x) \in \{0, 1\} \forall x \in \mathbb{X}$

In this case, define Rejection Region: $R_\phi = \{x \in \mathbb{X} : \phi(x) = 1\}$

In this case, define Acceptance Region: $A_\phi = \{x \in \mathbb{X} : \phi(x) = 0\}$

Type I Error: $P_\theta(\text{Reject } H_0), \theta \in \Theta_0$

Type II Error: $1 - P_\theta(\text{Reject } H_0), \theta \in \Theta_1$

Note: Probability I reject:

$P_\theta(\text{Reject } H_0) = \mathbb{E}[\phi(X)] =: \beta_\phi(\theta)$

Here, $\beta_\phi(\theta)$ is called the power function.

Define: Let ϕ be a test function

1. The size of ϕ is $\underset{\theta \in \Theta_0}{sup} \beta_\phi(\theta)$

2. ϕ is a level α test if its size is no more than α

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Example: $X_1, ..., X_5 \sim Exp(\theta)$

$H_0 : \theta = 1$ vs. $H_1 : \theta < 1$

$$\phi(x) = \begin{cases} 1, & \bar{X} < 1 \\ 0, \bar{X} \geq 1 \end{cases}$$

What is the power function?

$\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(x)] = P_\theta(\bar{X} < 1)$

Since $\bar{X} \sim Gamma(n, \theta/n), \chi^2_\nu \stackrel{D}{=} Gamma(\frac{\nu}{2}, 2) \implies \frac{2n\bar{X}}{\theta} \sim \chi^2_{2n} \implies \beta_\phi(\theta) = P(\chi^2_{10} < \frac{10}{\theta})$

Size of ϕ: $\beta_\phi(1) = P(\chi^2_{10} < 10) \approx 0.56$

Type II Error: What is the probability of a Type II Error if $\theta = 1/3$?

$1 - \beta_\phi(1/3) = P(\chi^2_1 0 > 30) \approx 0.001$

16.1 Most Powerful Tests

For a particular H_0, define for $\alpha \in (0,1), \Phi_\alpha = \{\phi : \phi \text{ is a test function with size } \leq \alpha\}$

Define: A test $\phi^* \in \Phi_\alpha$ is said to be (uniformly) most powerful (MP - Simple H_0 or UMP - Composite H_0) for $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ if $\beta_{\phi^*}(\theta) \geq \beta_\phi(\theta) \forall \phi \in \Phi_\alpha$ and all $\theta \in \Theta_1$.

Example: $X_i \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ known

$H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1, \theta 1 > \theta_0$

The z-tests has test function: $\phi(x) = I(\bar{X} > \theta_0 + z_{1-\alpha} * \frac{\sigma}{\sqrt{n}}) = I(\bar{X} > \theta_0 + k)$

To enforce a size contstraint:

$\beta_\phi(\theta_0) = P_{\theta_0}(\bar{X} > \theta_0 + k) = P(\frac{\bar{X}-\theta_0}{\sigma/\sqrt{n}} > \frac{k}{\sigma/\sqrt{n}}) = 1 - \Phi(\frac{k}{\sigma/\sqrt{n}})$, Φ is the $N(0,1)$ CDF

Want this to be $\leq \alpha \implies \Phi(\frac{k}{\sigma/\sqrt{n}}) \geq 1 - \alpha \implies \frac{k}{\sigma/\sqrt{n}} \geq z_{1-\alpha} \implies k \geq z_{1-\alpha}\frac{\sigma}{\sqrt{n}}$

$\beta_\phi(\theta_1) = P_{\theta_1}(\bar{X} > \theta_0 + k) = P(Z > \frac{\theta_0 - \theta_1 + k}{\sigma/\sqrt{n}}) = 1 - \Phi(\frac{\theta_0-\theta_1+k}{\sigma/\sqrt{n}})$

To get a required size, we can pick any k as highlighted above. However, power goes down as k increases. Thus, to maximize power under H_1 and satisfy the level constraint, $k = \frac{\sigma}{\sqrt{n}} z_{1-\alpha}$

Theorem (Neyman-Pearson Lemma):

Consider testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ (both are simple). Then any test of the form (with $k > 0, \gamma \in [0,1]$):

$$\phi(x) = \begin{cases} 1, & f(x|\theta_1) > kf(x|\theta_0) \\ \gamma, & f(x|\theta_1) = kf(x|\theta_0) \\ 0, & f(x|\theta_1) < kf(x|\theta_0) \end{cases}$$

is an MP test of a level equal to its size.

Proof: $\frac{f(x|\theta_1)}{f(x|\theta_0)}$ has a continuous distribution (the likelihood ratio).

Let $\alpha = \beta_\phi(\theta_0)$ and $\tilde{\phi}$ be any other test function with $\beta_{\tilde{\phi}}(\theta_0) \leq \alpha$

$$\begin{aligned}
\beta_\phi(\theta_1) - \beta_{\tilde{\phi}}(\theta_1) &\geq \beta_\phi(\theta_1) - \beta_{\tilde{\phi}}(\theta_1) - k[\beta_\phi(\theta_0) - \beta_{\tilde{\phi}}(\theta_0)] \\
&= \mathbb{E}_{\theta_1}[\phi(x) - \tilde{\phi}(x)] - k\,\mathbb{E}_{\theta_0}[\phi(x) - \tilde{\phi}(x)] \\
&= \int\limits_{\mathbb{X}} [\phi(x) - \tilde{\phi}(x)] f(x|\theta_1) dt - k \int\limits_{\mathbb{X}} [\phi(x) - \tilde{\phi}(x)] f(x|\theta_0) dx \\
&= \int\limits_{A_\phi} -\tilde{\phi}(x)[f(x|\theta_1) - kf(x|\theta_0)]dx + \int\limits_{R_\phi} [1-\phi(x)][f(x|\theta_1) - kf(x|\theta_0)]dx \\
&\geq 0
\end{aligned}$$

Example: (normal revisited)

$f(x|\theta) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum(x_i-\theta)^2}$

$\frac{f(x|\theta_1)}{f(x|\theta_0)} = e^{\frac{n\bar{X}(\theta_1-\theta_0)}{\sigma^2} - \frac{n(\theta_1^2-\theta_0^2)}{2\sigma^2}} = c_1 e^{c_2\bar{X}}$, $(c_1, c_2 > 0)$

The test $\phi(x) = \begin{cases} 1, & c_1 e^{c_2\bar{X}} > k \\ 0, & c_1 e^{c_2\bar{X}} < k \end{cases}$ is an MP test relative to other tests of smaller size.

There exists $\tilde{k}$ (1-1 function with k) such that $\phi(x) = \begin{cases} 1, & \bar{X} > \tilde{k} \\ 0, & \bar{X} < \tilde{k} \end{cases}$

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Example: $X_i \overset{\text{iid}}{\sim} Bern(\theta), H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1, 0 < \theta_0 < \theta_1 < 1$

$\frac{f(x||\theta_1)}{f(x|\theta_0)} = \frac{\theta_1^{\sum x_i}(1-\theta_1)^{n-\sum x_i}}{\theta_0^{\sum x_i}(1-\theta_0)^{n-\sum x_i}} = \left(\frac{1-\theta_1}{1-\theta_0}\right)^n \left(\frac{\frac{\theta_1}{1-\theta_1}}{\frac{\theta_0}{1-\theta_0}}\right)^{\sum x_i}$

The ratio within the exponent of $\sum x_i$ is monotonically increasing (due to $\theta_1 > \theta_0$, etc.)

$\implies$ the NP test is: $\begin{cases} 1, & \frac{f(x|\theta_1)}{f(x|\theta_0)} > k \\ \gamma, & \frac{f(x|\theta_1)}{f(x|\theta_0)} = k \\ 0, & \frac{f(x|\theta_1)}{f(x|\theta_0)} < k \end{cases} = \begin{cases} 1, & \sum x_i > c \\ \gamma, & \sum x_i = c \\ 0, & \sum x_i < c \end{cases}$

Take $n = 2, c = 1, \theta_0 = 1/2, \theta_1 = 3/4, \gamma = 0$

$\beta_\phi = P_\theta(\sum x_i > 1) \implies P_{\theta_0}(\sum x_i > 1) = \theta_0^2 = 1/4 = size(\phi)$

$\implies P_{\theta_1}(\sum x_i > 1) = \theta_1^2 = 9/16 = 1 - P(\text{Type II Error})$

Find c such that the size is $\alpha > 0.3$

$\beta_\phi(\theta) = P_\theta(\sum x_i > c) + \gamma P_\theta(\sum x_i = c) \implies \beta_\phi(1/2) = P_{1/2}(\sum x_i > c) + \frac{\gamma}{4}\binom{2}{c} I_{c\in\{0,1,2\}}$

$P_{1/2}(\sum x_i > c) \begin{cases} 3/4, & c = 0 \\ 1/4, & c = 1 \\ 0, & c = 2 \end{cases}$

Need to choose c such that the first piece is as close to, but not exceeding, α.

Take $c = 1$ and set $\gamma = \frac{\alpha - P_{1/2}(\sum x_i > 1)}{1/4(2)} = 2(0.3 - 1/4) = .6 - .5 = .1$

<u>In Class Example</u>: $X_i \overset{\text{iid}}{\sim} Pois(\theta), H_0 : \theta = \theta_0, H_1 : \theta = \theta_1, 0 < \theta_0 < \theta_1$

$\frac{f(x|\theta_1)}{f(x|\theta_0)} = \frac{\theta_1^{\sum x_i} e^{-n\theta_1}}{\theta_0^{\sum x_i} e^{-n\theta_0}}$ (the factorials cancel) $= c(\frac{\theta_1}{\theta_0})^{\sum x_i}$

Because $\theta_1 > \theta_0$, that function is monotonically increasing in terms of $\sum x_i$, so we can write $\phi(x) = \begin{cases} 1, & \sum x_i > t_0 \\ \gamma, & \sum x_i = t_0 \\ 0, & \sum x_i < t_0 \end{cases}$

Power function: $\beta_\phi(\theta) = P_\theta(\sum x_i > t_0) + \gamma P(\sum x_i = t_0)$

<u>Example</u>: $X_i \overset{\text{iid}}{\sim} Pois(\theta), H_0 : \theta = \theta_0, H_1 : \theta = \theta_1, 0 < \theta_1 < \theta_0$

$\frac{x|\theta_1}{f(x|\theta_0)} = c(\frac{\theta_1}{\theta_0})^{\sum x_i} \implies \phi(x) = \begin{cases} 1, & \sum x_i < t_0 \\ \gamma, & \sum x_i = t_0 \\ 0, & \sum x_i > t_0 \end{cases}$

Let $n = 10, \theta_0 = 3$, find t_0, γ such that ϕ has size 0.05

$\beta_\phi(\theta_0) = P_{\theta_0}(\sum x_i < t_0) + \gamma P_{\theta_0}(\sum x_1 = t_0)$

These actually correspond to n=1

$P_{\theta_0}(\sum x_i < 1) \approx 0.0497, P_{\theta_0}(\sum x_i < 2) \approx 0.1991$

Thus we choose $t_0 = 1$ (remember, highest value where that prob is less than α), $\implies \gamma = \frac{\alpha - 0.0497}{e^{-30} 30^1} \approx 0.002$

<u>Example</u>: $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), H_0 : \theta = \theta_0, H_1 : \theta = \theta_1, \theta_1 > \theta_0$

$\phi(x) = I(\bar{X} > \theta_0 + \frac{\sigma}{\sqrt{n}} Z_{\alpha - 1})$. This test is MP for H_0 vs. H_1 (note that it does not depend on θ_1). It is also uniformly MP (UMP) for $H_0 : \theta = \theta_0, H_1' : \theta > \theta_0$

Recall that: $\beta_\phi(\theta) = 1 - \Phi(\frac{\sqrt{n}(\theta_0 - \theta)}{\sigma} + z_{1-\alpha}) \implies \underset{\theta \le \theta_0}{sup} \beta_\phi(\theta) = \alpha \implies$ It is UMP for $H_0' : \theta \le \theta_0, H_1' : \theta > \theta_0$

<u>Example</u>: $X_1, X_2 \sim Unif(\theta, \theta + 1), H_0 : \theta = 0, H_1 : 0 < \theta < 1$

$\frac{f(x|\theta_1)}{f(x|\theta_0)} = \frac{I(\theta_1 < x_{(1)} < x_{(2)} < \theta_1 + 1)}{(\theta_0 < x_{(1)} < x_{(2)} < 1)} = \begin{cases} 1, & \theta_1 < x_{(1)} < x_{(2)} < 1 \\ 0, & 0 < x_{(1)} < \theta_1 \\ \infty, & x_{(2)} > 1 \end{cases}$

Take $k = 0$: the power function is $\beta_\phi(\theta_0) = (1 - \theta_1)^2 + \gamma(1 - (1 - \theta_1)^2)$

Take $k = 1$: the power function is $\beta_\phi(\theta_0) = \gamma(1 - \theta_1)^2$

Since the rejection region always depends on θ_1, the NP test for a fixed θ_1 can't be UMP.

If I restrict $H_1' : 0 < \theta < 1 - \sqrt{\alpha}$, then the UMP test is is: $\phi^*(x) = \begin{cases} 1, & X_{(2)} > 1 \text{ or } X_{(1)} > 1 - \sqrt{\alpha} \\ 0, & \text{otherwise} \end{cases}$

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Suppose T is a sufficient statistic:

$\frac{f(x|\theta_1)}{f(x|\theta_0)} = \frac{h(x)g(T(x);\theta_1)}{h(x)g(T(x);\theta_0)} = \frac{g(T(x);\theta_1)}{g(T(x);\theta_0)} >, =, < k$

Definition: Let T be any univariate statistic and $g(t|\theta)$ be its PDF/PMF. We say $g(t|\theta)$ has the monotone likelihood ratio (MLR) property if: $\frac{g(t|\theta_2)}{g(t|\theta_1)}$ is non-decreasing in t whenever $\theta_2 > \theta_1$.

Example: $X_i \overset{\text{iid}}{\sim} Gamma(\theta, scale = \beta), \beta$ is known.

The sufficient statistic $T = \sum ln(x_i)$. $f(x|\theta) = \frac{1}{(\Gamma(\theta)\beta^\theta)^n} e^{(\theta-1)T - \frac{\sum x_i}{\beta}} = e^{-\frac{1}{\beta}\sum x_i} * \frac{1}{\Gamma^n(\theta)\beta^{n\theta}} e^{(\theta-1)T}$

Let $\theta_2 > \theta_1 \implies \frac{g(t|\theta_2)}{g(t|\theta_1)} = \left[\frac{\Gamma(\theta_1)}{\Gamma(\theta_2)}\right]^n \beta^{-n(\theta_2-\theta_1)} e^{(\theta_2-\theta_1)t} \implies T = \sum ln(x_i)$ has the MLR property.

Theorem - (Karlin-Rubin): $\Theta \in \mathbb{R}^1$

Let $f(x|\theta)$ be a family with univariate sufficient statistic T that has MLR. Then any test function of the form $\phi(x) = \begin{cases} 1, & T(X) > t_0 \\ \gamma, & T(X) = t_0 \\ 0, & T(X) < t_0 \end{cases}$ is the UMP test of level $\gamma = P_{\theta_0}(T > t_0)$ for $H_0 : \theta \leq \theta_0, H_1 : \theta > \theta_0$.

Proof: Let $\bar{\phi}$ be any test function with size at most α.

Step 1: Show that $\underset{\theta \leq \theta_0}{sup} \beta_\phi(\theta) = \alpha$. See Problem 8.26(c) (it's long)

Step 2: Consider $H_0' : \theta = \theta_0, H_1' : \theta = \theta', \theta' > \theta_0$. Then, $T(x) >, =, < t_0 \implies \frac{g(T(x)|\theta')}{g(T(x)|\theta_0)} >, =, < k' \implies \frac{f(x|\theta')}{f(x|\theta)} >, =, k'$

Therefore for any test function ϕ' satisfying $\beta_{\phi'} \leq \beta_\phi(\theta_0) = \alpha$, $\beta_{\phi'}(\theta') \leq \beta_\phi(\theta')$

Since θ' is arbitrary, ϕ must be UMP for $H_0' : \theta = \theta_0, H_1 : \theta > \theta_0$

Step 3: $\beta_{\bar{\phi}} \leq \underset{\theta \leq \theta_0}{sup} \beta_{\bar{\phi}}(\theta) \leq \alpha$

By step 2 $\implies \beta_{\bar{\phi}}(\theta') \leq \beta_\phi(\theta')$ for any $\theta' > \theta_0$

Example: $X_i \overset{\text{iid}}{\sim} Unif(0, \theta)$

$T = X_{(n)}$ is sufficient. Let $\theta_2 > \theta_1 > 0, H_0 : \theta \leq \theta_0, H_1 : \theta > \theta_0$

$\frac{f(x|\theta_2)}{f(x||\theta_1)} = \frac{\theta_2^{-n}}{I(X_{(n)} \leq \theta_2)} \theta_1^{-n} I(X_{(n)} \leq \theta_1) = \begin{cases} (\frac{\theta_1}{\theta_2})^n, & X_{(n)} \leq \theta_1 \\ \infty, & \theta_1 < X_{(n)} \leq \theta_2 \end{cases}$ which is a non-decreasing function.

Since $T = X_{(n)}$ has MLR, any test of the form $\phi(x) = \begin{cases} 1, & X_{(n)} > t_0 \\ 0, & X_{(n)} < t_0 \end{cases}$ is UMP of its size.

Set the Type I Error Rate to α and find t_0.

$\beta_\phi(\theta_0) = P_{\theta_0}(X_{(n)} > t_0) = 1 - P_{\theta_0}(X_1 < t_0)^n = \begin{cases} 1, & t_0 > \theta_0 \\ 1 - (\frac{t_0}{\theta_0})^n, & t_0 < \theta_0 \end{cases} \implies 1 - (\frac{t_0}{\theta_0})^n = \alpha \implies t_0 = \theta_0(1-\alpha)^{1/n}$

If you switch the conditions on the hypostheses, you switch the order of the test function.

Defaults: MLR non-decreasing, $H_0 : \theta \leq \theta_0$

H_0/MLR	Non-Dec	Non-Inc
$\theta \leq \theta_0$	$T > t_0$	$T < t_0$
$\theta \geq \theta_0$	$T < t_0$	$T > t_0$

indicates when we reject.

Example: (Non-existence of UMP for two-sided tests)

$X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ known.

Let $\phi^+(x)$ be UMP for $H_0^+ : \theta \leq \theta_0, H^+1_1 : \theta > \theta_0$, and let $\phi^-(x)$ be UMP for $H_0^- : \theta \geq \theta_0, H_1^- L\theta < \theta_0$ for level α.

Now suppose ϕ is UMP for level α for $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$

Note that $\beta_{\phi^+}(\theta_0) = \beta_{\phi^-}(\theta_0) = \alpha \implies \beta_\phi(\theta') \leq \beta_{\phi^+}(\theta'), \theta' > \theta_0$ because both ϕ and ϕ^+ are size α for H_0 vs. H_1^+

Similarly, $\beta_\phi(\theta') \leq \beta_{\phi^-}(\theta'), \theta' < \theta_0$. By NP, $\phi = \phi^+ = \phi^-$ (contradiction since $\phi^- \neq \phi^+$).

19 Likelihood Ratio Tests

$H_0 : \theta \in \Theta_0, H_1 : \theta \in \Theta_1, \Theta = \Theta_0 \cup \Theta_1$

Definte: The (generalized) Likelihood ratio is $\lambda(x) = \frac{\underset{\theta\in\Theta_0}{sup}\ L(\theta|x)}{\underset{\theta\in\Theta}{sup} L(\theta|x)}$

Define $\hat{\theta}$ to be the ordinary MLE and $\hat{\theta}_0 = \underset{\theta\in\Theta_0}{argmax} L(\theta|x)$. Then $\lambda(x) = \frac{L(\hat{\theta}_0|x)}{L(\hat{\theta}|x)}$

Example: $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ known.

$H_0 : \theta \leq \theta_0, H_1 : \theta > \theta_0$

We know $\hat{\theta} = \bar{X}$. $l'(\theta|x) = \frac{n}{\sigma^2}(\bar{x} - \mu) = \begin{cases} < 0, & \theta > \bar{x} \\ = 0, & \theta = \bar{x} \\ > 0, & \theta < \bar{x} \end{cases}$

$\implies \hat{\theta}_R$ (Restricted MLE, MLE under Θ_0) $= \begin{cases} \bar{X}, & \bar{X} \leq \theta_0 \\ \theta_0, & \bar{X} > \theta_0 \end{cases}$

$$\lambda(x) = \frac{L(\hat{\theta}_R)|x}{L(\hat{\theta}|x)} = \begin{cases} 1, & \bar{X} \leq 0 \\ \frac{e^{-\frac{1}{2\sigma^2}\sum(x_i - \theta_0)^2}}{e^{-\frac{1}{2\sigma^2}\sum(x_i - \bar{X})^2}}, & \bar{X} > \theta_0 \end{cases}$$

Reject if $\lambda(x) < c, 0 < c < 1$

Simplifying: $ln(\lambda(x)) < ln(c) \iff \frac{-n}{2\sigma^2}(\bar{X} - \theta_0)^2 < ln(c)$

Reject if $\bar{X} > \theta_0$ (because the likelihood ratio indicates we only don't reject when $\bar{X} > \theta_0$) and $\bar{X} - \theta_0 > \sqrt{\frac{-2\sigma^2}{n} ln(c)} = k \implies$ Reject when $\bar{X} > \theta_0 + k$. This is basically just the Karlin-Rubin.

Example:

Same as above, σ^2 is known, but now, $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$

For the likelihood ratio, no maximiazation under null because it is simple, and we already did that part above.

We know $ln(\lambda(x)) = \frac{-2\sigma^2}{n}(\bar{X} - \theta_0)^2$. When is this $< ln(c)$

$\frac{-2\sigma^2}{n}(\bar{X} - \theta_0)^2 < ln(c) \iff |\bar{X} - \theta_0| > \sqrt{\frac{-2\sigma^2}{n} ln(c)} \iff \bar{X} > \theta_0 + k$ or $\bar{X} < \theta_0 - k$

In fact, this is UMP unbiased

Example

Now, σ^2 unknown

$H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$ (θ is now the full paramater space, both θ and σ^2)

$\hat{\theta} = \bar{X}, \hat{\sigma}^2 = \frac{n-1}{n} S^2$

$\hat{\theta}_R = \theta_0, \hat{\sigma}^2_R = \frac{1}{n} \sum (x_i - \theta_0)^2$

$\lambda(x) = \frac{(2\pi\hat{\sigma}^2_R)^{-n/2} e^{-\frac{1}{2\hat{\sigma}^2_R} \sum (x_i - \theta_0)^2}}{(2\pi\hat{\sigma}^2)^{-n/2} e^{-\frac{1}{2\hat{\sigma}^2} \sum (x_i - \bar{X})^2}}$

$-\frac{1}{2\hat{\sigma}^2_R} \sum (x_i - \theta_0)^2 = \frac{-1}{2(1/n)} = \frac{-n}{2}$

$S^2 = \frac{1}{n} \sum (x_i - \bar{X})^2$

$\implies \lambda(x) = (\frac{\hat{\sigma}^2_R}{\hat{\sigma}^2})^{-n/2} = a_n(1 + \frac{(\bar{X} - \theta_0)^2}{S^2})^{-n/2}$

When $T = \frac{\bar{X} - \theta_0}{S/\sqrt{n}}, \lambda(x) = a_n(1 + \frac{T^2}{n})^{-n/2} \implies \lambda(x) < c \iff |T| > t_0$

$\implies$ the Likelihood ratio test (LRT) is the two-sided t-test

For size α, we want: $P_{\theta_0}(|T| > t_0) = \alpha \implies t_0 = qt(1 - \frac{\alpha}{2}, df = n - 1)$

Example $X_i \overset{\text{iid}}{\sim} Unif(0, \theta), \theta > 0, H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$

$\lambda(x) = \frac{L(\theta_0|x)}{L(X_{(n)}|x)} = \frac{\theta_0^{-n} I(X_{(n)} \leq \theta_0)}{X_{(n)}^{-n} I(X_{(n)} \leq X_{(n)})} = (\frac{X_{(n)}}{\theta_0})^n I(X_{(n)} \leq \theta_0) = \begin{cases} (\frac{X_{(n)}}{\theta_0})^n, & X_{(n)} \leq \theta_0 \\ 0, & X_{(n)} > \theta_0 \end{cases}$

We reject when $X_{(n)} > \theta_0$ or if $X_{(n)} < \theta_0$ and $(\frac{X_{(n)}}{\theta_0})^n < c$

$\alpha = P_{\theta_0}(X_{(n)} \leq \theta_0 c^{1/n}) = (P_{\theta_0}(X_1 < \theta_0 c^{1/n}))^n = (\frac{\theta_0 c^{1/n}}{\theta_0 - 0}) = c$

$\implies \phi(x) = I(\lambda(x) < \alpha) = \begin{cases} 1, & X_{(n)} > \theta_0 \text{ or } X_{(n)} < \theta_0 \alpha^{1/n} \\ 0, & \text{otherwise} \end{cases}$

Example: $X_i \overset{\text{iid}}{\sim} Gamma(\gamma, scale = \theta), \gamma$ is known

$H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$

$\hat{\theta} = \frac{\sum x_i}{n\gamma} = \frac{1}{\gamma}\bar{X}$

$$
\begin{aligned}
\implies \lambda(x) &= \frac{L(\theta_0|x)}{L(\hat{\theta}|x)} \\
&= \frac{[\Gamma(\gamma)\theta_0^\gamma]^{-n} e^{-\frac{1}{\theta_0}\sum x_i}}{[\Gamma(\gamma)\hat{\theta}^\gamma]^{-n} e^{-\frac{1}{\hat{\theta}}\sum x_i}} \\
&= (\frac{\bar{X}}{\theta_0\gamma})^{n\gamma} e^{n\gamma[1-\frac{\bar{X}}{\theta_0\gamma}]} \\
&= h(\frac{\bar{X}}{\theta_0\gamma})^n \text{ where } h(y) = y^\gamma e^{\gamma(1-\gamma)} \\
\implies h'(y) &= \gamma y^{\gamma-1} e^{\gamma(1-y)} - \gamma y^\gamma e^{\gamma(1-y)} = \gamma y^{\gamma-1} e^{\gamma(1-y)}[1-y]
\end{aligned}
$$

This is increasing from $y \in (0,1)$, then decreasing from $y \in (1,\infty)$.

We want to find some c such that $h(y_1) = c$ and $h(y_2) = c$ and the total region between those points is α

To find a size α test of the form: $\phi(x) = \begin{cases} 1, & \bar{X} > b_2 \text{ or } \bar{X} < b_1 \\ 0, & \text{otherwise} \end{cases}$

We need:

(1) $P_{\theta_0}(\bar{X} > b_2) + P_{\theta_0}(\bar{X} < b_1) = \alpha$

(2) $h(\frac{b_1}{\theta_0\gamma}) = h(\frac{b_2}{\theta_0\gamma})$

System of equations, use root finding methods to find the cutoffs.

19.1 Large-Sample Testing

Example: $X_i \overset{\text{iid}}{\sim} N(\theta,\sigma^2)$ study 2 variable version for EXAM, σ^2 known, $H_0: \theta = \theta_0, H_1: \theta \neq \theta_0$

$\lambda(x) = e^{-\frac{n}{2\sigma^2}(\bar{X}-\theta_0)^2}$

$-2ln(\lambda(x)) = (\frac{\sqrt{n}(\bar{X}-\theta_0)}{\sigma})^2 = (N(0,1))^2 \overset{D}{=} \chi_1^2$

Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta), H_0: \theta = \theta_0, H_1: \theta \neq \theta_0$

$\lambda(x) = \frac{\frac{1}{\theta_0^n} e^{-\frac{n\bar{X}}{\theta_0}}}{\frac{1}{\bar{X}} e^{-n}} = e^{n[1-\frac{\bar{X}}{\theta_0}+ln(\frac{\bar{X}}{\theta_0})]}$

$-2ln(\lambda(x)) = 2n[g(\bar{X}) - g(\theta_0)], g(x) = \frac{x}{\theta_0} - ln(x)$

$g'(x) = \frac{1}{\theta_0} - \frac{1}{x} \implies g'(\theta_0) = 0$

Need to use the 2nd order Delta Method. Since $\sqrt{n}(\bar{X}-\theta_0) \xrightarrow{D} N(0,\theta_0^2)$, the 2nd order delta method says: $2n(g(\bar{X})-g(\theta_0)) \approx g''(\theta_0)[\sqrt{n}(\bar{X}-\theta_0)]^2 = [\frac{\sqrt{n}(\bar{X}-\theta_0)}{\theta_0}]^2 \xrightarrow{D} \chi_1^2$

$$\phi(x) = \begin{cases} 1, & \lambda(x) < c \\ 0, & \lambda(x) \geq c \end{cases} = \begin{cases} 1, & -2ln(\lambda(x)) > k \\ 0, & -2ln(\lambda(x)) \leq k \end{cases}$$

Choosing $k = \chi^2_{1,1-\alpha} : \beta_\phi(\theta_0) \xrightarrow{D} \alpha$

<u>Theorem</u>: Wilks'

Under regularity assumptions, if $H_0 : \theta = \theta_0$ is true, $\Theta \subset \mathbb{R}^k$: $-2ln(\lambda(x)) \xrightarrow{D} \chi^2_k$.

<u>Proof</u>: (k=1)

$l(\theta|x) = l(\hat{\theta}|x) + (\theta - \hat{\theta})l'(\hat{\theta}|x) + \frac{(\theta-\hat{\theta})^2}{2} l''(\hat{\theta}|x) + R_n(\theta, \hat{\theta})$

The regularity conditions imply that $R_n(\theta_0, \hat{\theta}) \xrightarrow{P} 0$. We also found MLE setting first derivate equal to 0, so that part drops out as well.

$-2ln(\lambda(x)) = -2ln(\frac{L(\theta_0|x)}{L(\hat{\theta}|x)}) = -2[l(\theta_0|x) - l(\hat{\theta|}x)] = 2[l(\hat{\theta}|x) - l(\theta_0|x)] = (-\frac{1}{n} l''(\hat{\theta}|x))[\sqrt{n}(\hat{\theta} - \theta_0)]^2 - R_n(\theta_0, \hat{\theta})$

1) $-\frac{1}{n} l''(\hat{\theta}|x) \xrightarrow{P} I_1(\theta_0)$

2) $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N(0, I_1^{-1}(\theta_0))$

$\implies -2ln(\lambda(x)) = \left[(-\frac{1}{n} l''(\hat{\theta}|x))^{1/2} \sqrt{n}(\hat{\theta} - \theta_0)\right]^2 - R_n(\theta_0, \hat{\theta}) \xrightarrow{D} [\sqrt{I_1(\theta_0)} N(0, I_1^{-1}(\theta_0))]^2 = [N(0,1)]^2 = \chi^2_1$

<u>Notes</u>:

1) Wilks' original theorem only works for simple Null hypotheses

2) Intuitively, I should just be able to reject if i) $|\hat{\theta} - \theta_0|$ is large or ii) $|\psi(x; \theta_0)|$ is large

19.2 Expanded Wilks' Theorem

Remember Wilks' Theorem: For $H_0 : \theta = \theta_0, \Theta_0 \in \mathbb{R}^k : -2ln(\lambda(x)) \xrightarrow{D} \chi^2_k$

<u>Theorem</u>: Wilks' Theorem part 2

$X_i \overset{\text{iid}}{\sim} f(x|\theta), H_0 : \theta \in \Theta_0, H_1 : \theta \notin \Theta_0, \Theta_0 = \{\theta \in \mathbb{R}^k : \theta_j = \theta_{0,j}, j = 1, ..., r\}$

Then, under $H_0 : -2ln(\lambda(x)) \xrightarrow{D} \chi^2_r$

<u>Example</u>: $X_i \overset{\text{iid}}{\sim} Beta(\alpha, \beta)$

$H_0 : \alpha = \beta, H_1 : \alpha \neq \beta$

Let $\theta_1 = \alpha - \beta, \theta_2 = \beta$

Now, $H_0 : \theta_1 = 0, H_1 : \theta_1 \neq 0$

$\hat{\alpha}_R = argmax L(\alpha, \alpha|X)$ is the null-constrained MLE. Remeber the null asumes $\alpha = \beta$

$\lambda(x) = \frac{L(\hat{\alpha}_R, \hat{\alpha}_R|X)}{L(\hat{\alpha}, \hat{\beta}|X)}$

Then $-2ln(\lambda(x)) \xrightarrow{D} \chi^2_1$

In this example, a natural test statistic might also be based on $T(x) = \hat{\alpha} - \hat{\beta}$

In this case, $\sqrt{n}(\hat{\alpha} - \hat{\beta}) \xrightarrow{D}_{H_0} N(0, V_0)$

We know, $\sqrt{n}(\begin{pmatrix}\hat{\alpha}\\ \hat{\beta}\end{pmatrix} - \begin{pmatrix}\alpha\\ \beta\end{pmatrix}) \xrightarrow{D} N(0, I^{-1}(\alpha,\beta))$

Take $g(x,y) = x - y$:

$$\begin{aligned}\sqrt{n}(\begin{pmatrix}\hat{\alpha}\\ \hat{\beta}\end{pmatrix} - \begin{pmatrix}\alpha\\ \beta\end{pmatrix}) &= \sqrt{n}(\hat{\alpha} - \hat{\beta} - (\alpha - \beta)) \\ &= \text{(taking null to be true) } \sqrt{n}(\hat{\alpha} - \hat{\beta} - 0) \\ &\xrightarrow{D} N(0, (1,-1)I_1^{-1}(\alpha,\alpha)\begin{pmatrix}1\\ -1\end{pmatrix})\end{aligned}$$

In general, this is what Wald tests do.

The Wald Statistic is $Z_w = \sqrt{nI_!(\hat{\theta})}(\hat{\theta} - \theta_0)$

$Z_w = \frac{\sqrt{n}(\hat{\theta}-\theta)}{\sqrt{1/I_1(\hat{\theta})}} \frac{\xrightarrow{D}}{\xrightarrow{P}} \frac{N(0, I_1^{-1}(\theta))}{I_1^{-1/2}(\theta)} = N(0,1)$

The Wald test rejects when $|Z_w| > Z_{1-\alpha/2}$

Another test statistic is the score statistic based on: $\psi(x;\theta) = \frac{\partial}{\partial\theta} ln(f(x|\theta))$

When we proved normality of MLEs, we showed: $\sqrt{n}\psi(x;\theta_0) \xrightarrow{D} N(0, I_1(\theta_0))$

The score test uses: $Z_s = \frac{\psi(x;\theta_0)}{\sqrt{nI_1(\theta_0)}}$

Example: $X_i \overset{\text{iid}}{\sim} Bern(\theta), H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$

Wald test: $\hat{\theta} = \bar{X}$ is the MLE, $I_1(\theta) = \frac{1}{\theta(1-\theta)}$

$Z_w = \sqrt{\frac{n}{\hat{\theta}(1-\hat{\theta})}}(\hat{\theta} - \theta) = \frac{\sqrt{n}(\bar{X}-\theta_0)}{\sqrt{\bar{X}(1-\bar{X})}}$

Score test: $\psi(x;\theta) = \frac{n(\bar{X}-\theta)}{\theta(1-\theta)}$

$Z_s = \frac{n(\bar{X}-\theta_0)}{\theta_0(1-\theta_0)\sqrt{\frac{n}{\theta_0(1-\theta_0)}}} = \frac{\sqrt{n}(\bar{X}-\theta_0)}{\sqrt{\theta_0(1-\theta_0)}}$

Hypothesis Testing

1) Definitions: H_0, test functions (ϕ), power functions, size, level, etc.

2) Uniformly Most Powerful

- Neyman-Pearson (simplie H_0, H_1)

- Karlin-Rubin (one-sided w/ MLR)

- Beyond that no guarantee for UMP

3) Likelihood Ratio Tests

- Reframe the test $\phi(x) = I(\lambda(x) < c)$ as $\phi(x) = I(T(X) < b_1) + I(T(X) > b_2)$

4) Asymptotic Tests

- Wilks' theorem

- Wald/Score Tests

20 Confidence Intervals

Definition: An interval estimate for θ is a pair of statistics $(L(X), U(x))$ such that $L(X) \leq U(X)$ almost surely.

Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta), \hat{\theta} = \bar{X}$

Example intervals:

$L(X) = \bar{X} - c, U(X) = \bar{X} + d; c, d > 0$

$L(X) = \frac{\bar{X}}{a}, U(X) = b\bar{X}; a, b > 1$

Definition: The Coverage Probability of an interval estimate $(L(X), U(X))$ at $\theta \in \Theta$ is $P_\theta(L(X) \leq \theta \leq U(X))$

Example: (cont'd)

$P_\theta(\bar{X} - c \leq \theta \leq \bar{X} + d) = P_\theta(\theta - d \leq \bar{X} \leq \theta + c), \bar{X} \sim Gamma(n, \frac{\theta}{n})$. Coverage depends on θ.

For the other interval:

$P_\theta(\frac{\bar{X}}{a} \leq \theta \leq b\bar{X}) = P_\theta(\frac{\theta}{b} \leq \bar{X} \leq a\theta) = P_\theta(\frac{1}{b} \leq \frac{\bar{X}}{\theta} \leq a), \frac{\bar{X}}{\theta} \sim Gamma(n, \frac{1}{n})$

Definition: For an interval estimate $(L(x), U(x))$, its confidence coefficient is: $\underset{\theta}{inf} P_\theta(L(x) \leq \theta \leq U(x))$

Note: We say an interval estimate is level $100(1-\alpha)\%$ if its confidence coefficient is atleast $1 - \alpha$.

Example: $X_i \overset{\text{iid}}{\sim} U(0, \theta)$

$T = X_{(n)}$ Some options for condfidence interval:

$L_1(x) = X_{(n)}, U_1(x) = X_{(n)} + c$

$L_2(x) = X_{(n)}, U_2(x) = aX_{(n)}, a > 1$

$P_\theta(L_1(x) \leq \theta \leq U_1(x)) = P_\theta(X_{(n)} \leq \theta \leq X_{(n)} + c) = P_\theta(\theta - c \leq X_{(n)} \leq \theta) = P_\theta(\theta - c \leq X_{(n)}) = 1 - P_\theta(X_{(n)} \leq \theta - c) = 1 - (1 - \frac{c}{\theta})^n$

Confidence Coefficient: $\lim_{\theta \to \infty} P_\theta(L_1(x) \leq \theta \leq U_1(x)) = 0$

$P_\theta(L_2(x) \leq \theta \leq U_2(x)) = P_\theta(aX_{(n)} \geq \theta) = P_\theta(X_{(n)} \geq \frac{\theta}{a}) = 1 - (\frac{1}{a})^n = 1 - \frac{1}{a^n}$

Confidence Coefficient is constant! Does not depend on θ

If I want a coefficient of $1 - \alpha$, set: $1 - \alpha = 1 - \frac{1}{a^n} \implies a = \alpha^{-1/n}$

20.1 Pivotal Intervals

Definition: A random variable $Q(x, \theta)$ is said to be pivotal (or a pivotal quantity) if its distribution does not depend on θ.

Location: $f(x|\theta) = g(x - \theta)$

Pivots: $X_i - \theta$, then any function of that value: $\bar{X} - \theta$, $\tilde{X} - \theta$ (median), etc. ie. Any $h(X_1 - \theta, ..., X_n - \theta)$

Scale: $f(x|\theta) = \frac{1}{\theta} g(\frac{x}{\theta})$

Pivots: $\frac{X_i}{\theta}$, then any function of that: $\bar{x}\theta, \frac{X_{(i)}}{\theta}$, etc. Any $h(\frac{X_1}{\theta}, ..., \frac{X_n}{\theta})$

For lcoation scale, do the location centering first, then the scale (ie. $\frac{X_1-\theta}{\sigma}$)

Definition: A confidence set $C(x) \subset \mathbb{R}$ has level $1-\alpha$ is $P(\theta \in C(x)) \geq 1-\alpha$

If $Q(X,\theta)$ is a pivot, then $C(x) = \{\theta : a \leq Q(x,\theta) \leq b\}; a < b$ forms a confidence set.

Why? Let D be the CDF of $Q(x,\theta)$. Then $P_\theta(\theta \in C(x)) = P_\theta(a \leq Q(x,\theta) < b) = D(b) - D(a)$

To ensure $C(x)$ has level $1-\alpha$ choose $\alpha_1, \alpha_2 > 0$ such that $\alpha_1 + \alpha_2 = \alpha; a = D^{-1}(\alpha_1), b = D^{-1}(1-\alpha_2)$

Typically, $Q(x,\theta)$ is decreasing in θ (for any fixed x). This means that for $Q(x)$ between a and b, there are corresponding values for $\theta : L(x), U(x)$

Example $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ is known

$Q(x,\theta) = \bar{x} - \theta \sim N(0, \frac{\sigma^2}{n})$

$a = \frac{\sigma}{\sqrt{n}}\Phi^{-1}(\alpha_1), b = \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1-\alpha_2)$

Solve $\bar{X} - L(x) = \Phi^{-1}(\frac{\sqrt{n}(1-\alpha_2)}{\sigma}), \bar{x} - U(x) = \Phi^{-1}(\frac{\sqrt{n}\alpha_1}{\sigma}) \implies (L(x), U(x)) = (\bar{X} - \frac{\sigma}{\sqrt{n}}Z_{1-\alpha_2}, \bar{X} - \frac{\sigma}{\sqrt{n}}Z_{\alpha_1})$

Example: $X_i \overset{\text{iid}}{\sim} N(0, \theta)$

$Q(x,\theta) = \frac{\sum x_i^2}{\theta} \sim \chi^2_n$

$a = \chi^2_{n,\alpha_1}, b = \chi^2_{n,1-\alpha_2} \implies \frac{\sum x_i^2}{L(x)} = \chi^2_{n,1-\alpha_2}, \frac{\sum x_i^2}{U(x)} = \chi^2_{n,\alpha_1}$

$\implies (L(x), U(x)) = \left(\frac{\sum x_i^2}{\chi^2_{n,1-\alpha_2}}, \frac{\sum x_i^2}{\chi^2_{n,\alpha_1}}\right)$

Example: $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Task for us: how do we build an interval on μ that doesn't depend on σ, etc.?

Pivot for μ: $\frac{\bar{x}-\mu}{S/\sqrt{n}} \sim t_{n-1}$

Pivot for σ^2: $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

For next class: For $Y \sim F_Y$ continuous, $U = F_Y(Y) \sim Unif(0,1)$. This is a pivot.

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Recall that, if T is some statistic with a continuous CDF $F_T(t|\theta)$, then $U = F_T(t|\theta) \sim U(0,1)$

Choose $Q(x,\theta) = F_T(T|\theta)$ as a pivot.

$C(x) = \{\theta : a \leq Q(x,\theta) \leq b\}$ is a confidence set with coefficient $b-a$ for $0 \leq a < b \leq 1$

Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta)$

Start with $T = \bar{x} \sim Gamma(n, \frac{\theta}{n})$

We could theoretically divide by θ to get a Gamma which does not depend on θ. Assuming we don't know that, we'll use the CDF method.

Define $F_T(t|\theta) = pgamma(t, n, \frac{\theta}{n})$

To get a level $1-\alpha$ CI, find all θ such that $\alpha_1 \leq F_T(\bar{x}|\theta) \leq 1-\alpha_2$, with $\alpha_1 + \alpha_2 = \alpha; \alpha_1, \alpha_2 > 0$

Example: $X_i \overset{\text{iid}}{\sim}$ with PDF $f(x|\theta) = e^{-(x-\theta)}, x > \theta$.

In this case, θ is a lower bound on x. Due to that, a reasonable estimator is $X_{(1)}$. You could also say that this is a location family, so a pivot would be $x - \theta$.

$T = X_{(1)} \overset{D}{=} \theta + Y_{(1)} m Y_i \overset{\text{iid}}{\sim} Exp(1) \implies F_T(t|\theta) = 1 - e^{-n(t-\theta)}$

Set α_1, α_2 before and solve: $\alpha_1 \leq 1 - e^{-n(t-\theta)} \leq 1 - \alpha_2 \implies C(x) = \{\theta : \alpha_1 \leq F_T(X_{(1)}|\theta) \leq 1 - \alpha_2\} = [x_{(1)} + \frac{1}{n} ln(\alpha_2), X_{(1)} + \frac{1}{n} ln(1-\alpha_1)]$

21.1 Inverting Tests

Given a confidence set $C(x)$, we can define a test function $\phi(x) = \begin{cases} 1, & \theta_0 \notin C(x) \\ 0, & \theta_0 \in C(x) \end{cases}$ for $H_0 : \theta = \theta_0$.

The size is $P_{\theta_0}(\theta_0 \notin C(x)) = 1 - P_{\theta_0}(\theta_0 \in C(x))$

We can also operate in the reverse order

Example: $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ is known.

The UMP unbiased size α test has acceptance region (for $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$), $A(\theta_0) = \{x : |\bar{x} - \theta_0| \leq \frac{\sigma}{\sqrt{n}} Z_{\alpha/2}\}$

We know $P_{\theta_0}(x \notin A(\theta_0)) = \alpha$ (true for any θ_0)

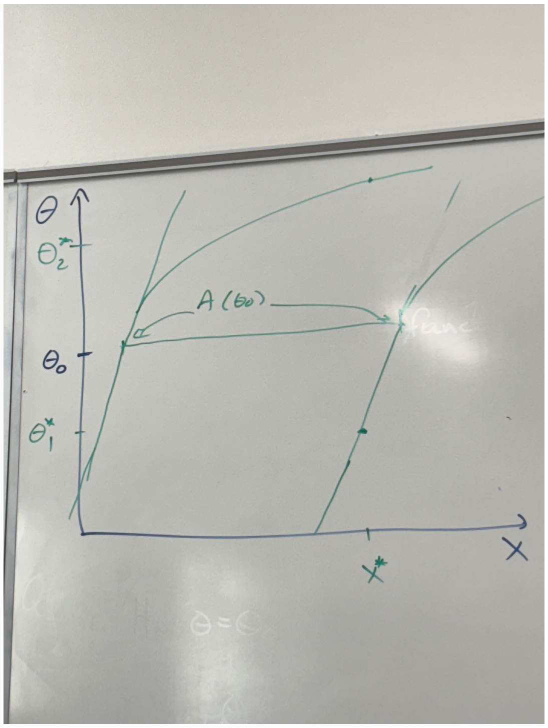
Then we can construct $C(x) = \{\theta_0 : x \in A(\theta_0)\}$

The coverage probability is $P_{\theta_0}(\theta_0 \in C(x)) = P_{\theta_0}(x \in A(\theta_0)) = 1 - \alpha$ (for any θ_0)

Theorem For each $\theta_0 \in \Theta$, let $A(\theta_0)$ be the acceptnace region of a size α test of $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$.

Then $C(x) = \{\theta_0 : x \in A(\theta_0)\}$ is a level $100(a - \alpha)\%$ confidence set.

Example illustration of what the inverse test is doing:



Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta)$

A size α LRT will accept when $\frac{\theta_0 e^{\frac{-n\bar{x}}{\theta_0}}}{\bar{x}e^{-n}} = \left[(\frac{\bar{x}}{\theta_0}e^{1-\frac{\bar{x}}{\theta_0}})\right]^n > c$

In fact, $A(\theta_0) = \{x : a \leq \bar{x} \leq b\}, P_{\theta_0}(a \leq \bar{x} \leq b) = 1-\alpha$, $\frac{a}{\theta_0}e^{1-\frac{a}{\theta_0}} = \frac{b}{\theta_0}e^{1-\frac{b}{\theta_0}}; a = a(\theta_0), b = b(\theta_0)$

$\implies C(x) = \{\theta_0 : x \in A(\theta_0)\} = \{\theta_0 : a(\theta_0 \leq \bar{x} \leq b(\theta_0))\}$

21.2 Length Minimization

Example $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \sigma^2$ known

Inverting the UMP inbiased test, $(\bar{X} - \frac{\sigma^2}{\sqrt{n}}Z_{1-\alpha/2}, \bar{X} + \frac{\sigma^2}{\sqrt{n}}Z_{1-\alpha/2})$

Using the pivot method on $\bar{X} - \theta$, $(\bar{X} - \frac{\sigma^2}{\sqrt{n}}Z_{1-\alpha/2}, \bar{X} + \frac{\sigma^2}{\sqrt{n}}Z_{1-\alpha/2}); \alpha_1 + \alpha_2 + \alpha$

To choose α_1, α_2, we can try to minimize the (expected) length .

In this example: $L = L(\alpha_1, \alpha_2) = \frac{\sigma}{\sqrt{n}}(Z_{1-\alpha_2} - Z_{\alpha_1})$

I need to minimize under the constraint that $\alpha_1 + \alpha_2 = \alpha$

Rerwite: $a = Z_{\alpha_1}, b = Z_{1-\alpha_2}$, so that $\Phi(a) = \alpha_1, \Phi(b) = 1 - \alpha_2 \implies \Phi(a) + 1 - \Phi(b) = \alpha \implies \Phi(b) - \Phi(a) = 1 - \alpha$

The Langrangian is $h(a, b, \lambda) = b - a + \lambda\{\Phi(b) - \Phi(a) - (1-\alpha)\}$ (the last term is just our condition written such that it equals 0)

$\frac{\partial h}{\partial a} = -1 - \lambda\phi(a)$

$\frac{\partial h}{\partial b} = 1 + \lambda\phi(b)$

$\frac{\partial h}{\partial \lambda} = \Phi(b) - \Phi(a) - (1-\alpha)$

Setting the first two to 0 implies $1 + \lambda\phi(a) = 1 + \lambda\phi(b) \implies \phi(a) = \phi(b)$ since $\lambda \neq 0$. This means $b = \pm a$.

Now, for the third equation to be 0, $\Phi(b) - \Phi(a) = 1 - \alpha; (a < \alpha < 1)$

We must have $a = -b = Z_{\alpha/2}$ since $\Phi(b) = \Phi(-a) = 1 - \Phi(a)$

Example: $X_i \overset{\text{iid}}{\sim} Exp(\theta)$

Pivot (scale family): $\frac{\bar{x}}{\theta} \sim Gamma(n, \frac{1}{n})$

$C(x) = \{\theta : q_{\alpha_1} \leq \frac{\bar{x}}{\theta} \leq q_{1-\alpha_2}\}; \alpha_1 + \alpha_2 = \alpha \implies \left[\frac{\bar{x}}{q_{1-\alpha_2}}, \frac{\bar{x}}{q_{\alpha_1}}\right]$

The length is $\bar{x}(\frac{1}{q_{\alpha_1}} - \frac{1}{q_{1-\alpha_2}})$. Note that this depends on $\bar{x}$ so we can't just minimize this because that would depend on θ which we don't want.

We can minimize $\mathbb{E}_\theta(Length)$ at each θ by minimizing the function $g(a,b) = \frac{1}{a} - \frac{1}{b}$ subject to the contrsaint $F(b) - F(a) = 1 - \alpha$ where F is the $Gamma(n, \frac{1}{n})$ CDF.

Lagrangian: The part we want to minimize, plus/minus (you can use either plus/minus because that part will just be 0)λ times the constraint (where the constraint is written as being equal to 0).

Set $h(a,b,\lambda) = [\frac{1}{a} - \frac{1}{b}] + \lambda[F(b) - F(a) - (1-\alpha)]$

$\frac{\partial h}{\partial a} = -\frac{1}{a^2} - \lambda f(a) \implies \lambda = \frac{1}{a^2 f(a)}$

$\frac{\partial h}{\partial b} = \frac{1}{b^2} + \lambda f(b) \implies \lambda = \frac{1}{b^2 f(b)}$

$\implies a^2 f(a) = b^2 f(b)$

$\frac{\partial h}{\partial \lambda} = F(b) - F(a) - (1-\alpha)$

So we have a two piece system of equations:

$a^2 f(a) - b^2 f(b) = 0$

$F(b) - F(a) - (1-\alpha) = 0$

Given an α, we can use a variety of techniques to solve this for a and b, such as root finding (R).

21.3 Large Sample Intervals

When n is large it can be beneficial to use:

1) Asymptotic Pivots

2) Invert large samples test

Wald Intervals

For testing $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$,

$Z_w = \frac{\hat{\theta} - \theta_0}{Std.Error(\hat{\theta})} \underset{H_0}{\overset{D}{\to}} N(0,1)$

This gives us an acceptance region of $A(\theta_0) = \{x : |Z_w| \leq Z_{1-\alpha/2}\}$

The corresponding confidence set is: $C(x) = \{\theta_0 : |Z_w| \leq Z_{1-\alpha/2}\} = \{\theta_0 : |\frac{\hat{\theta}-\theta_0}{SE(\hat{\theta})}| \leq Z_{1-\alpha/2}\} = [\hat{\theta} - Z_{1-\alpha/2}SE(\hat{\theta}), \hat{\theta} + Z_{1-\alpha/2}SE(\hat{\theta})]$

Remember that this is the asymptotic, so as n gets larger, the test approaches the $N(0,1)$ and approaches the size α test.

Note: $Q(x,\theta) = \frac{\hat{\theta}-\theta}{SE(\hat{\theta})} \underset{\text{large n}}{\overset{D}{\approx}} N(0,1)$

This leads to a large sample pivotal interval: $[\hat{\theta} - SE(\hat{\theta})Z_{1-\alpha/2}, \hat{\theta} - SE(\hat{\theta})Z_{\alpha_1}]$

Score-based Intervals

To test $H_0 : \theta = \theta_0, H_1 : \theta = \theta_0$, use:

$Z_s = \frac{\psi(x;\theta)}{\sqrt{nI_1(\theta_0)}} \underset{H_0}{\overset{D}{\rightarrow}} N(0,1)$

Either using this as an asymptotic pivot or inverting an asymptotic size α test (the score test) gives:

$C(x) = \{\theta_0 : |\frac{\psi(x;\theta_0)}{\sqrt{nI_1(\theta_0)}}| \leq Z_{1-\alpha/2}\}$

This is usually an interval, but often difficult to solve exactly.

LR Intervals

For testing $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$, $-2ln(\lambda(x)) \underset{H_0}{\overset{D}{\rightarrow}} \chi^2_1$ (for one parameter).

Test-based confidence set: $C(x) = \{\theta_0 : -2ln(\lambda(x)) \leq \chi^2_{1-\alpha}\}$ (θ_0 is in numerator of $\lambda(x)$)

The pivot-based confidence set is $C(x) = \{\theta_0 : \chi^2_{a,\alpha_1} \leq -2ln(\lambda(x)) \leq \chi^2_{1,1-\alpha_2}\}$

Transformation-based Interval

Attempt to avoid estimating the standard error for Wald methods.

For MLEs, the standard error is: $Var\hat{\theta} \approx \frac{1}{nI_1(\theta)} \approx \frac{1}{nI_1(\hat{\theta})}$. In this case we'd say $SE(\hat{\theta}) = \sqrt{\frac{1}{nI_1(\hat{\theta})}}$

If $I_1(\theta)$ is not known, we can use $I\hat{I}_1(\theta) = -\frac{1}{n}\sum\limits_{i=1}^{n} \frac{\partial^2}{\partial\theta^2} ln(f(x_i|\theta))$ so we could use $SE(\hat{\theta}) = \sqrt{\frac{1}{n\hat{I}_1(\hat{\theta})}}$

This can be messy and lead to poor coverage.

Recall: $\sqrt{n}(\hat{\theta} - \theta) \overset{D}{\rightarrow} N(0, \sigma^2) \implies \sqrt{n}(g(\hat{\theta}) - g(\theta)) \overset{D}{\rightarrow} N(0, \sigma^2_\theta\{g'(\theta)\}^2)$

We want g to satisfy $g'(\theta) = \frac{1}{\sigma_\theta} \implies g(\theta) = \int \frac{1}{\sigma_\theta} d\theta$

This is a monotonic (increasing) transformation $\implies C(x) = \{\theta : |\sqrt{n}(g(\hat{\theta}) - g(\theta))| \leq Z_{1-\alpha/2}\} = \{\theta : g(\hat{\theta}) - \frac{1}{\sqrt{n}}Z_{1-\alpha/2} \leq g(\theta) \leq g(\hat{\theta}) + \frac{1}{\sqrt{n}}Z_{1-\alpha/2}\}$

$\implies [g^{-1}(g(\hat{\theta}) - \frac{1}{\sqrt{n}}Z_{1-\alpha/2}), g^{-1}(g(\hat{\theta}) + \frac{1}{\sqrt{n}}Z_{1-\alpha/2})]$

g is called a variance stabilizing transformation for $\hat{\theta}$